

Temporary Premium Benefits on a Multi-Channel Platform: Effects on Engagement and Monetization

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Abstract

Firms use temporary premium benefits to acquire, retain, and monetize consumers. In this paper, we study how these promotions affect user behavior on a multi-channel social livestreaming platform. We collect novel user-level panel data that capture viewership, chat engagement, and subscription purchases across popular creator channels. To estimate causal effects, we use quasi-exogenous variation in the allocation of premium benefits together with double/debiased machine learning. During the benefit month, temporary premium benefits increase watch time by more than three hours on the promoted channel and more than four hours on other channels. The promotion initially cannibalizes subscriptions to the promoted channel, but it generates positive spillover effects to other channels and eventually increases subscriptions to the promoted channel as well. Subscription spillovers concentrate on channels similar to the promoted channel, and are strongest among users without a subscription history, consistent with learning about the value of premium benefits. We use our treatment effect estimates within a multi-objective optimization framework to investigate how platforms can target promotional access to temporary premium benefits to balance different, potentially competing, objectives.

Keywords: causal inference, machine learning, digital marketing, media consumption, targeting, policy evaluation

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1 Introduction

Firms use a wide range of promotional strategies to acquire, retain, and monetize consumers. A common feature of many of these strategies is that they provide consumers temporary access to products, services, or features that would otherwise command a premium. For example, a video-streaming service may offer a free trial of its subscription plan, an airline loyalty program may provide a complimentary seat upgrade, or a generative AI platform may temporarily upgrade users from a basic plan to a premium tier with expanded access to advanced models and features. Such promotions may affect subsequent behavior through several mechanisms: they can provide consumers with information about product quality or fit ([Akerberg, 2001](#); [Bawa and Shoemaker, 2004](#); [Foubert and Gijsbrechts, 2016](#); [Li et al., 2019](#)), influence consumer habits ([Dubé et al., 2010](#); [Liu-Thompkins and Tam, 2013](#); [Shah et al., 2014](#)), and affect brand loyalty ([Anderson and Simester, 2004](#); [Lewis, 2004](#); [Wang et al., 2016](#)).

Despite their prevalence, these promotions, which we refer to as temporary premium benefits, can be difficult to evaluate and learn from. Participation can be driven by consumers' underlying interest in the promoted product, as when users enroll in a free trial. Eligibility for the promotion can be driven by targeting rules that favor consumers expected to be especially valuable or responsive, as with loyalty rewards, complimentary upgrades, or targeted premium access. Finally, effects can be context-specific, such as when the promotion is delivered around a new content release or a product launch. These sources of selection mean that a comparison of recipients to non-recipients can easily conflate the promotion's causal effect with the characteristics of the users and contexts in which it was delivered.

Even when its causal effects can be identified, the managerial value of a promotion may depend on how broadly its effects are evaluated. For example, a promotion may increase engagement while reducing purchases of the promoted product. These effects may also change over time, particularly when short-run responses differ from long-run outcomes ([Mela et al., 1997](#); [Pauwels et al., 2002](#)). For firms offering multiple products or services, the effects

may extend beyond the promoted product through spillovers to other parts of the portfolio (Sahni et al., 2017; Patabhiraiah et al., 2019). Evaluations that focus on a single outcome or time horizon may therefore miss important consequences of the promotion.

In this paper, we study three research questions. First, how do temporary premium benefits causally affect engagement, monetization, and retention? Second, how do these effects vary across products, consumers, contexts, and time horizons, and what mechanisms explain them? Third, how should firms allocate temporary premium benefits when the most responsive consumers differ across different managerial objectives?

We answer these questions using a novel dataset that tracks user behavior across creator channels on Twitch, a large social livestreaming platform. On this platform, users can watch live content, participate in chat, and purchase channel-specific subscriptions that provide premium benefits. We study a platform feature that algorithmically allocates temporary access to subscription benefits, helping us address several typical obstacles to causal inference. First, because recipients do not opt in or out, users cannot self-select into the promotion. Second, we observe recipients, comparable non-recipients, and a rich set of pre-allocation behavior, allowing us to model the quasi-random allocation algorithm. Third, we observe many allocation events, meaning our estimated effects are not conflated with any single channel or context.

We use channel-specific audience information to define treated users and comparable control users for each allocation event. Because the allocation algorithm’s decision logic is not observed, we use double/debiased machine learning (Chernozhukov et al., 2018) to model treatment assignment and user outcomes as a function of pre-treatment behavior. We estimate the effects of temporary premium benefits on viewing, chat engagement, paid subscriptions, and retention, distinguishing behavior on the channel where the benefit is received from behavior on all other observed channels. We measure these outcomes during the benefit month and in the three months after the benefits expire, allowing us to distinguish contemporaneous effects from longer-term changes in behavior. We then explore treatment-

effect heterogeneity and the role of different mechanisms in explaining behavior. Finally, we use causal forests (Athey et al., 2019) to estimate user-level treatment effects, incorporate them into a multi-objective framework (Rafieian et al., 2025), and evaluate counterfactual targeting policies.

Temporary access to premium benefits leads to substantial changes in user behavior, with the direction and magnitude of those changes dependent on both the outcome and the time horizon. During the benefit month, recipients watch more than three additional hours on the channel where they receive premium benefits and more than four additional hours on other channels, with increased viewing persisting after benefits expire. The promotion also increases chat usage, indicating that these promotions can increase social engagement. Subscription behavior effects are more nuanced. Recipients are initially slightly less likely to be subscribed to the channel where benefits are received in the month immediately following the benefit period, consistent with short-run cannibalization, but this effect reverses to a significant increase in the following month. At the same time, recipients are significantly more likely to subscribe to other channels on the platform, suggesting the effects of premium benefits can spill over to other consumption contexts. Temporary premium benefits also increase retention (decrease churn) in the months following the benefit period. Together, these results demonstrate that an evaluation based only on immediate conversion on the promoted channel would reach a substantially different conclusion from one that incorporates cross-channel behavior and longer-run outcomes.

The cross-channel spillovers for paid subscriptions vary systematically across channels and users. Spillovers are largest for channels with content that is similar to the channel where benefits were received. They are also most prevalent among users who engage relatively less with the channel on which benefits were received compared to other channels. This pattern suggests that providing benefits outside a user’s preferred channel may be a useful strategy for generating subscription spillovers. This pattern is strongest among users without a subscription history, who often respond to the promotion by purchasing a subscription to

their most-watched channel. This finding is consistent with users learning the value of a subscription and then subscribing where that value is highest.

More broadly, the users with the largest treatment effects on trial-channel subscriptions, other-channel subscriptions, and retention are not the same users. Consequently, no single user segment or simple targeting rule simultaneously maximizes multiple platform objectives. We use causal forests to estimate user-level heterogeneous treatment effects in order to evaluate different targeting policies. The platform’s observed allocation performs no better than random targeting, while policies weighted towards specific objectives substantially improve the outcomes they are targeted to maximize. However, maximizing one objective comes with significant opportunity costs in terms of sacrificing potential improvements in other objectives. For example, a policy that weights the three outcomes equally achieves more than 80% of the estimated benefit under the corresponding single-objective policy on each outcome. These findings show that the value of firm-allocated promotions depends not only on identifying consumers with large causal responses, but also on explicitly accounting for which responses the firm values when deciding whom to target.

This research makes three contributions to our understanding of how firms should evaluate and allocate temporary premium benefits. First, we extend prior research on measuring promotional effects by studying a setting in which users receive temporary premium benefits without having to opt in. We show that an evaluation of the promotion based on immediate conversions on the promoted channel would detect short-run cannibalization but miss both the later subscription gains on that channel and the positive spillovers elsewhere on the platform. Second, we document how promotional spillovers vary across channels and users. Subscription spillovers concentrate on channels with content similar to the promoted channel and are strongest among users with no prior subscription history. These patterns are consistent with consumers learning the value of premium benefits through the promotion, and they suggest that multi-product firms that internalize spillover effects across their portfolio may find a promotion worthwhile even when it cannibalizes sales of the promoted

product. Finally, we quantify the opportunity costs of single-objective targeting when firms value multiple outcomes. Small additional gains on one objective can require substantial sacrifices on others, while policies that balance objectives can retain much of the value of targeting on each individual outcome.

The remainder of the paper is structured as follows. [Section 2](#) reviews the relevant literature. [Section 3](#) describes the empirical context and the temporary premium benefit promotion we study. [Section 4](#) describes the data and sample construction. [Section 5](#) presents the identification strategy and estimation approach. [Section 6](#) presents the average effects of temporary premium benefits across outcomes and time horizons, and investigates the mechanisms driving spillovers. [Section 7](#) evaluates single- and multi-objective targeting policies. [Section 8](#) concludes.

2 Literature

This research builds on several literatures, including how premium benefits affect paid conversion and user behavior, the spillover effects of promotional interventions, and strategies for targeting promotions to achieve different outcomes.

A large body of literature explores how exposure to premium benefits through free trials, freemium models, and paywalls affects user behavior. This research addresses two related questions. First, what drives adoption of premium benefits? [Bapna and Umyarov \(2015\)](#) find that peer influence increases premium conversion in a freemium service, while [Oestreicher-Singer and Zalmanson \(2013\)](#) find that social and content engagement are associated with premium conversion. [Yoganarasimhan et al. \(2023\)](#) demonstrate that shorter free trial duration can increase customer acquisition, retention, and profitability. [Reza et al. \(2021\)](#) find that prior usage levels are important predictors of the take-up and effects of free samples of experience goods. Second, how does access to premium benefits change user behavior? [Iyengar et al. \(2022\)](#) find that retail membership programs increase purchase frequency and

basket size, while [Datta et al. \(2018\)](#) show that music streaming access increases consumption quantity and diversity. Our research contributes to both streams by studying how temporary access to premium benefits affects engagement and consumption, which in turn influences subscription adoption. While most previous studies analyze contexts where users self-select into the promotion, one exception is [von Wangenheim and Bayón \(2007\)](#), which studies the impact of unexpected service upgrades. This research extends these findings by studying a multi-service digital platform where users receive promotions without opting in or selecting the upgraded service.

For multi-product firms, promotional interventions can generate spillover effects that require careful management through strategic design and targeting. [Bawa and Shoemaker \(2004\)](#) show that free samples can increase sales among prior and new customers, but may also cannibalize planned purchases. [Sahni et al. \(2017\)](#) find that targeted offers on a ticket-resale platform increase spending on nonpromoted ticket categories. [Pattabhiramaiah et al. \(2019\)](#) show how firms with multiple distribution channels can internalize cross-channel effects. When firms are not positioned to internalize spillover effects, competing firms can free ride on these benefits ([Shapiro, 2018](#)). [Chae et al. \(2017\)](#) show that seeded word-of-mouth campaigns redirect attention across products, increasing target product discussions while decreasing competitor discussions. We contribute to this literature by quantifying spillovers from temporary premium access on a digital platform. The spillovers are largest for products similar to the promoted one, and among consumers who have never purchased before—those with the most to learn about the product’s value. The implication for targeting is that temporary access is most effective when it lands close to what the consumer already prefers, where there is no existing sale to cannibalize.

Prior research demonstrates the tradeoffs associated with different promotional targeting strategies and the importance of measuring heterogeneous responses. [Datta et al. \(2015\)](#) highlight tradeoffs between customer acquisition and quality, finding that trial-acquired customers often have lower lifetime value than customers from other channels. [Foubert and](#)

Gijsbrechts (2016) document the “double-edged” nature of free trials. Free trials accelerate adoption but may attract lower-quality customers. Ascarza (2018) and Lemmens and Gupta (2020) challenge conventional targeting approaches, showing that targeting high churn-risk customers (where the stakes are highest) can be suboptimal compared to targeting based on estimated treatment sensitivity (where there are the greatest predicted gains). Yoganarasimhan et al. (2023) evaluate personalized targeting policies that assign different targeting treatments based on individual-level predictions of a single outcome of interest. Yang et al. (2024) show that firms can target on long-term outcomes, such as long-term revenue, by predicting them from customers’ short-term outcomes. We extend this targeting literature by applying the multi-objective framework of Rafeian et al. (2025) to show how platforms can balance gains across outcomes when allocating temporary premium benefits.

Finally, this research contributes to a growing literature on social livestreaming platforms, where the variety of observable behaviors and contexts creates unique environments for studying media consumption behaviors and the effects of promotional interventions (Lin et al., 2021; Lu et al., 2021; Forderer et al., 2026; Simonov et al., 2023; Huang and Morozov, 2025; Chaudhry et al., 2025).

3 Empirical Context

3.1 Twitch

We study user behavior on Twitch,¹ a popular social livestreaming platform that connects content creators (referred to as “streamers”), users (“viewers”), and advertisers. Twitch, like most livestreaming platforms, helps creators monetize their content through ads and subscription revenue.

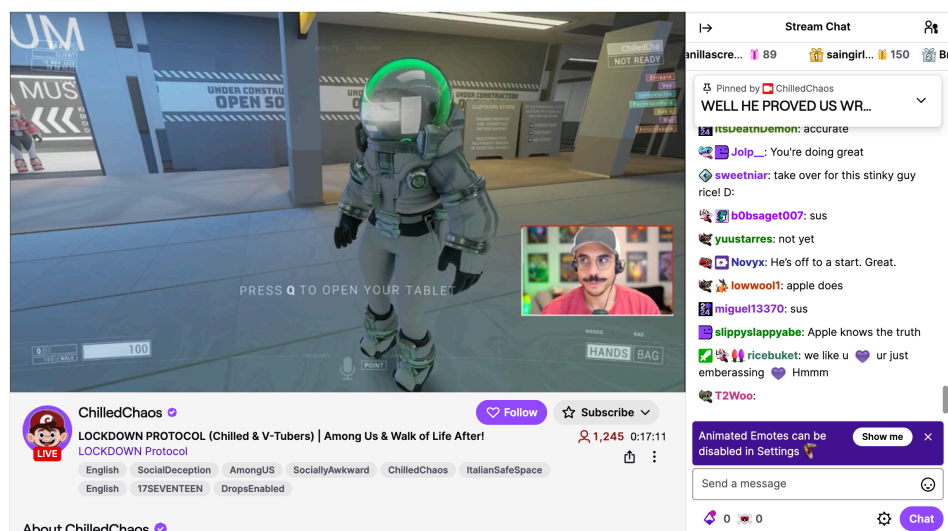
Each creator operates their own channel on the platform. Creator content often focuses on video games, but includes a wide range of topics such as music, politics, and conversations

¹<https://twitch.tv>

with the channel audience (“Just Chatting”). Figure 1 shows the typical user experience on a channel.

Users can subscribe to a creator’s channel, enabling channel-specific benefits including ad-free viewing (e.g., eliminating pre-roll ads) and special chat privileges (e.g., badges and emojis). Users can subscribe to these benefits in one of three ways. They can self-subscribe for a fee (paid subscription) or by linking their Amazon Prime membership (Prime subscription). Users can also receive gift subscriptions from other users. Gift subscriptions offer the same benefits to the recipient as self-subscriptions. Users gift subscriptions to other users for many reasons: to financially support creators, to strengthen a creator’s community, to engage in a random act of kindness. Users can gift as few as one or up to one hundred subscriptions at a time to a creator’s audience. All subscriptions last for 30 days and cost about \$4.99.²

Figure 1: Twitch User Experience



Notes: Screen capture of a typical user experience when viewing a creator’s channel. The user can see streamed video content, a webcam view of the creator, a chat panel, and information on the current stream (creator’s name, stream content tags, time since start of stream, user subscription status, number of viewers).

²Prices vary based on device and region. More details on gift subscriptions are available at https://help.twitch.tv/s/article/gift-subscriptions?language=en_US.

3.2 Temporary Premium Benefits

When a user purchases more than one gift subscription at a time, an algorithm, rather than the purchaser, determines which users receive subscriptions. Twitch describes the algorithm as follows (emphasis added):

We use an algorithm to help us select gift recipients *starting with eligible viewers* in chat, then followers, mods, and other *factors that identify members of a community*. Our algorithm also *avoids giving trolls subs*. We are constantly improving our algorithm to detect this behavior.

While these subscriptions share features with free trials and gifts, they differ in several ways. Like a free trial, these subscriptions give recipients limited-time access to the same benefits as standard subscriptions at no cost. However, unlike a free trial, recipients neither sign up to receive benefits nor do they provide payment details or commit to recurring charges after the initial benefit period ends.³ Additionally, despite the platform label “gift,” the algorithmic, impersonal nature of allocation distances this context from conventional gift-giving dynamics.⁴ We therefore refer to this subset of gift subscriptions as *allocated* subscriptions, and the benefits they confer as temporary premium benefits.⁵

We are interested in estimating the effects of temporary premium benefits on user behavior. While the platform’s algorithm generates quasi-random variation in allocation, we do not observe the algorithm’s decision rules. A naïve approach might treat this context as an experiment with uniformly random allocation, in which case the effect of receiving the benefits could be estimated by comparing the post-treatment behaviors of users who received treatment to those that did not receive it. As evidenced by the quote above, allocation cor-

³Einav et al. (2025) show how passive enrollment and recurring charges can influence behavior in subscription contexts via inattention and switching costs.

⁴It has been shown that intentionality plays an important role in how people interpret and respond to others’ behavior (Clark, 1996). Moreover, recipients attribute less agency and intentionality to algorithmic decisions, particularly for tasks involving human skills, such as gift-giving, leading to reduced emotional reactions (Lee, 2018).

⁵Figure A.2 shows an example announcement of a user purchasing forty allocated subscriptions that are distributed across the channel’s community.

relates with user behaviors. Moreover, the algorithm may balance stated objectives, e.g., avoiding giving trolls the benefits, with unstated objectives, e.g., maximizing future profit.

This allocation algorithm has several implications for causal estimation. First, if observed characteristics capture the relevant confounders and we accurately model the allocation process, the residual variation in allocation can be as good as random. Second, if the algorithm prioritizes certain behaviors (e.g., chatting) or avoids users with particular characteristics (e.g., users that already subscribe, or users with bot-like behaviors), it will be important for us to measure behaviors and characteristics that are strongly correlated with those determinants of allocation.

4 Data

In this section, we discuss the data, treatment, and sample construction. We define pre-treatment user behaviors that are important for determining allocation and explaining variation in the effects of receiving temporary premium benefits. Lastly, we motivate and define the outcomes we analyze.

4.1 Data Collection

We combine data from multiple sources to build an expansive view of user behavior on the Twitch platform. Our analysis focuses on viewership and engagement with the top 100 English-language channels. These channels cover a range of content, including gaming, e-sports, music streaming, and chatting. We collect viewership, chat engagement, and subscription information across the 100 streamers for a nine-month period spanning July 2022 to March 2023. [Online Appendix A](#) lists all creators, explains selection criteria, and details the data collection process.

We collect user-level viewing behavior for each channel, recorded at approximately five-minute intervals. These viewing records allow us to measure watch time per channel, the

frequency of viewing sessions, and the number of distinct channels watched within different time frames.⁶

We supplement viewership data with the transcripts of channel-specific chat feeds. Chat feeds record every message in a creator’s channel, including user, timestamp, and message content. Chat feeds also record channel announcements, including when any user purchases a subscription for themselves or other users.

4.2 Sample

Our unit of analysis is an *eligible* user in a *cohort*. A user is *eligible* for an allocated subscription if they are not currently subscribed to the channel, and are observed on the channel within a six-minute window prior to the allocation event.⁷ A *cohort* is the set of all eligible users for a given allocation event. Eligibility is an important criterion that restricts the sample to users who could have received temporary premium benefits. We do not compare eligible users to ineligible users—those who watched the same channel at a different time or who watched other channels—as these users may differ in systematic ways.

Within a cohort, eligible users are *treated* if they receive temporary premium benefits; all other eligible users form a pool from which we sample *control* users. We include all eligible treated users in our sample. We sample control users from the pool of eligible users at a fixed 20:1 ratio of control users to treated users. This fixed sampling ratio ensures each treated user is equally informative while maintaining a sufficient control pool to estimate counterfactual behaviors. For example, if 40 eligible users received temporary premium benefits, the sample cohort would include the 40 treated users and a random sample of 800

⁶The analysis focuses on logged-in users, who comprise approximately 63% of channel audiences and demonstrate aggregate viewing patterns that closely mirror overall viewership (Simonov et al., 2023). We focus on logged-in users because their behaviors are observable and they represent the potential user base for subscription purchases and benefits.

⁷The algorithm does not award benefits to current subscribers. The five-minute frequency of data collection ensures at least one observation per channel in the six-minute window leading up to any cohort start time.

control users.⁸ Our final estimation sample comprises 136,424 treated and 2,726,139 control user-cohort observations across 13,089 cohorts.⁹

4.3 Pre-Treatment User Behaviors

Our objective is to understand how temporary premium benefits influence multiple measures of user engagement across different time horizons. We begin by outlining pre-treatment behaviors that are likely to predict allocation and that may explain variation in how users respond to temporary premium benefits.

To capture a broad scope of engagement, we distinguish between behavior on the channel where the user received the trial of premium benefits (the *trial* channel) and across all other observed channels (*other* channels). The composition of engagement may explain differences in behavioral responses. For example, users that engage more with other channels than the trial channel may have greater propensity for cross-channel spillovers. We define four categories of pre-treatment behaviors that may be useful in predicting outcomes: watch, chat, past subscription purchases, and overall platform activity. We rely on observed behaviors, which prior research has shown to be informative predictors of consumer responses (Matz and Netzer, 2017; Lemmens et al., 2026). We include multiple daily, weekly, and monthly lags of pre-treatment behavior to capture dynamics, such as state dependence and persistence in behavior (Keane, 1997; Seetharaman et al., 1999; Dubé et al., 2010), prior experience on the platform, and preference for variety (McAlister and Pessemier, 1982; Kim et al., 2002; Datta et al., 2018).

For every eligible user, *watch* behavior is defined as the number of minutes the user spends connected to a particular channel during a specified time period. *Chat* behavior is defined

⁸When sampling control users, we exclude ever-treated users—users who receive temporary premium benefits in any cohort in our data—from the control pool of every cohort. Ever-treated users’ post-treatment behaviors may be contaminated by their own promotion, making them invalid control comparisons for other treated users. Ever-treated users are 4.0% of users in our data.

⁹The ratio is exactly 20:1 in 99.87% of cohorts. In 17 cohorts (0.13%), such as cohorts that occur in the very beginning of a creator’s streaming session when the audience is small, there are too few eligible users to exactly maintain this ratio, for a total shortfall of 2,341 control users.

as the number of messages the user writes in a channel’s chat feed. *Subscription* behavior is defined as whether or not the user has an active paid subscription during a specified time period.¹⁰ Additionally, we measure several *platform* behaviors: session count, the number of times a user starts watching a channel during a specified time period, and channel variety, the number of unique channels accessed by the user.

The pre-treatment columns of [Table 1](#) present summary statistics for monthly behavior-channel-time observables that we use to predict allocation and model treatment response.¹¹ At the time of treatment, most eligible users are not subscribed to any channel—0% on the trial channel (by construction) and 1.9% across all other observed channels. A somewhat larger share held a subscription at some point in the preceding 30 days—1.6% on the trial channel and 2.6% across all other channels—implying that a small number of subscriptions lapsed shortly before treatment. During the month prior to treatment, users watch an average of 1,709 minutes (about 57 minutes per day) on the trial channel and 1,879 minutes (63 minutes per day) across all other observed channels. The distributions of watch and chat behaviors have large variance and are heavily right-skewed.

4.4 Outcomes

The outcomes we define closely parallel the pre-treatment behaviors. We continue to distinguish between behavior on the trial and other channels in order to determine which behaviors are changed by temporary premium benefits, and whether the total level of activity on the platform increases or is redistributed. Our panel also allows us to observe whether behavioral changes persist after the expiration of the benefit period, and whether these effects change in direction or magnitude over time. We define multiple monthly-level outcome periods, in-

¹⁰One-month subscriptions account for 98% of all paid subscriptions in our sample. However, there do exist paid subscriptions that last for two or more months. Therefore, we define a user’s subscription behavior in a particular period based on the time period in which the subscription was active, regardless of when the payment was made.

¹¹Additional daily-level watch and chat observables are described in [Table A.2](#).

cluding the entire benefit period ($M1$), and each of the three months after the benefit period expires ($M2$, $M3$, $M4$).

Table 1: Pre-Treatment Behaviors and Outcomes

Behavior	Channel	Months Pre		Months Post			
		M-2	M-1	M1	M2	M3	M4
Watch	Trial	1,335.07 (2,343.77)	1,708.91 (2,362.03)	1,765.32 (2,593.35)	1,214.98 (2,293.19)	1,069.22 (2,188.48)	778.67 (1,893.17)
	Other	1,890.57 (3,559.58)	1,878.55 (3,070.09)	1,915.64 (4,979.83)	1,687.21 (4,460.45)	1,578.78 (4,684.90)	1,193.75 (3,382.14)
Chat	Trial	10.58 (117.20)	13.06 (127.13)	13.26 (130.57)	9.86 (118.89)	9.05 (116.98)	8.32 (111.46)
	Other	8.72 (92.84)	9.41 (96.91)	8.92 (102.10)	7.89 (98.21)	7.49 (104.66)	6.94 (92.51)
Subscribe	Trial	0.022 (0.147)	0.016 (0.125)	0.066 (0.249)	0.023 (0.151)	0.023 (0.151)	0.025 (0.157)
	Other	0.023 (0.150)	0.026 (0.159)	0.028 (0.164)	0.024 (0.154)	0.022 (0.147)	0.021 (0.142)
Sessions	Trial	16.32 (23.94)	21.08 (24.55)	20.05 (25.10)	14.10 (21.89)	12.49 (21.15)	9.23 (18.72)
	Other	31.13 (51.06)	31.40 (46.53)	30.27 (54.08)	25.67 (49.96)	23.84 (57.84)	18.12 (42.37)
Variety	All	5.31 (4.94)	5.76 (4.76)	5.50 (4.59)	4.69 (4.43)	4.33 (4.36)	3.42 (4.09)
Engaged	All	0.902 (0.298)	1.000 (0.000)	0.997 (0.052)	0.899 (0.302)	0.854 (0.353)	0.748 (0.434)

Notes: Cells report means, with standard deviations in parentheses. M-2 and M-1 are pre-treatment months; M1–M4 are post-treatment months, with M1 being the benefit period. Subscribe-Trial in M1 includes allocated premium benefits. The sample contains 2,862,563 user-cohort observations across 13,089 cohorts. Daily covariate descriptives are provided in [Table A.2](#).

Outcomes are defined as above and grouped into the same four categories: watch, chat, future subscription purchases, and overall platform activity. Watch behavior is an important metric for both individual content creators and the platform, directly affecting advertising revenue, creator rankings, and growth metrics. Chat behavior is an important measure of engagement for both creators and the platform. Social interactions between viewers and creators enhance the live viewing experience and can indicate engaging or high-quality content

(Godes and Mayzlin, 2004). Subscriptions after the expiration of the benefit period help the platform measure the effectiveness of the promotion in converting non-paying users to paid users. There are many other behaviors the platform may care about that extend beyond engagement with a particular channel. We measure multiple post-treatment platform-level outcomes, including Channel Variety, the number of distinct channels a user watches during the outcome interval, Session Count, the number of separate viewing sessions a user initiates during that interval, and Engaged, an indicator for any watch or chat activity on an observed channel during the month. The post-treatment columns of Table 1 present summary statistics for monthly behavior-channel-time outcomes.

5 Empirical Strategy

This section presents the empirical strategy for estimating the causal effects of temporary premium benefits. We first outline the causal framework and identifying assumptions, then describe the estimation approach and its implementation.

5.1 Causal Framework

Our causal framework follows the canonical potential outcomes framework (Rubin, 1974). Let $Y_{ij}(W_i)$ denote the realization of outcome j should user i receive premium benefits ($W_i = 1$) or not ($W_i = 0$). The causal effect of the benefits on user i 's behavior is given by $Y_{ij}(1) - Y_{ij}(0)$, and the average treatment effect (ATE) of the benefits on user behavior is $\tau_j \equiv \mathbb{E}[Y_{ij}(1) - Y_{ij}(0)]$. Incorporating pre-treatment user characteristics and behaviors X_i , the conditional average treatment effect is $\tau_j(x) \equiv \mathbb{E}[Y_{ij}(1) - Y_{ij}(0) \mid X_i = x]$.

Average treatment effects are identified under many potential sets of restrictions on the relationship between Y_j , W , and X , the simplest case being random assignment of W , together with a stable unit treatment value assumption. A weaker identifying assumption

is that of unconfoundedness: $(Y_j(0), Y_j(1)) \perp W \mid X$. That is, after controlling for observed characteristics, the assignment process does not depend on a user’s potential outcomes.

The stable unit treatment value assumption requires that the potential outcomes for any one user do not vary with the assignment of premium benefits to other users, i.e., there is no interference across users. Conditional treatment effects are identified under an additional sufficient overlap condition, $0 < P(W = 1 \mid X) < 1$, i.e., treatment assignment is never deterministic conditional on observables.

5.2 Double/Debiased Machine Learning

We use double/debiased machine learning (DML), with separate models for benefit allocation and user outcomes (Chernozhukov et al., 2018). The allocation model, $e(x)$, describes the probability that a user receives premium benefits given observed pre-treatment characteristics, x :

$$e(x) = \Pr(W_i = 1 \mid X_i = x). \quad (1)$$

Each outcome model, $m_j(x)$, describes expected behavior given the same characteristics:

$$m_j(x) = \mathbb{E}[Y_{ij} \mid X_i = x]. \quad (2)$$

Following Chernozhukov et al. (2018), we use the partialling-out form (Robinson, 1988) of DML. For each outcome j , our preferred estimator is the regression of the residuals from the outcome model, $Y_{ij} - \hat{m}_j(X_i)$, on the residuals from the allocation model, $W_i - \hat{e}(X_i)$:

$$\hat{\tau}_j = \frac{\sum_i [Y_{ij} - \hat{m}_j(X_i)] [W_i - \hat{e}(X_i)]}{\sum_i [W_i - \hat{e}(X_i)]^2}. \quad (3)$$

Under unconfoundedness, $\hat{\tau}_j$ is a consistent estimator of the overlap-weighted average treatment effect of Li et al. (2018). Overlap weighting gives less weight to users for whom

allocation is closer to deterministic. As a result, it avoids giving disproportionate influence to allocations among users who are extremely likely or unlikely to receive benefits.

Additionally, to study heterogeneity in treatment effects, we estimate conditional treatment effects using causal forests:

$$\hat{\tau}_j(x) = \frac{\sum_i \hat{\alpha}_{ij}(x) [Y_{ij} - \hat{m}_j(X_i)] [W_i - \hat{e}(X_i)]}{\sum_i \hat{\alpha}_{ij}(x) [W_i - \hat{e}(X_i)]^2}, \quad (4)$$

where $\hat{\alpha}_{ij}(x)$ are neighborhood weights that determine each observation’s contribution to the effect estimate at x . The weights are specific to outcome j , and the allocation and outcome models are as defined above. Causal forests provide consistent estimates of $\tau_j(x)$ under the identifying assumptions above and regularity conditions established in [Athey et al. \(2019\)](#).

5.3 Implementation

We use gradient-boosted trees to estimate a separate outcome model for each outcome, and a single allocation model. Although similar techniques (e.g., random forests, regression forests, neural networks) could also be used to estimate these models, we selected gradient-boosted trees for their predictive accuracy and flexibility in modeling nonlinear relationships. Each model is fitted on one half of the cohorts and used to predict for the other half, so each observation’s cohort is excluded from the model’s estimation sample. All outcomes are demeaned at the cohort level to mitigate context-specific effects. This demeaning subsumes creator, time, and creator-time level effects. Hyperparameter tuning and additional estimation details are described in [Online Appendix B](#).

While we do not observe the exact features used in the platform’s allocation algorithm, identification requires that we condition on the factors that influence both allocation and outcomes. The platform describes its algorithm as prioritizing viewers in chat using factors that identify members of a channel’s community. We include a broad set of watch, chat, subscription, and platform covariates that capture engagement with the channel and plat-

form that is closely related to these stated criteria. Prior engagement is also likely to predict post-treatment behavior, given persistence and state dependence in consumption (Keane, 1997; Dubé et al., 2010), so conditioning on these behaviors addresses the most plausible sources of confounding, which improves the precision of both average and conditional treatment-effect estimates. Moreover, because the estimator is based on Neyman-orthogonal scores, it is insensitive to small errors in the allocation and outcome models (Chernozhukov et al., 2018).¹² Any remaining confounder would bias our estimates only to the extent that it explains variation in allocation and outcomes beyond what these pre-treatment behaviors already capture. We assess identification assumptions and sensitivity to confounding alongside our presentation of the results in the next section.

6 Results

In this section, we assess the assignment model, present treatment effect estimates, investigate mechanisms driving important effects, and discuss robustness.

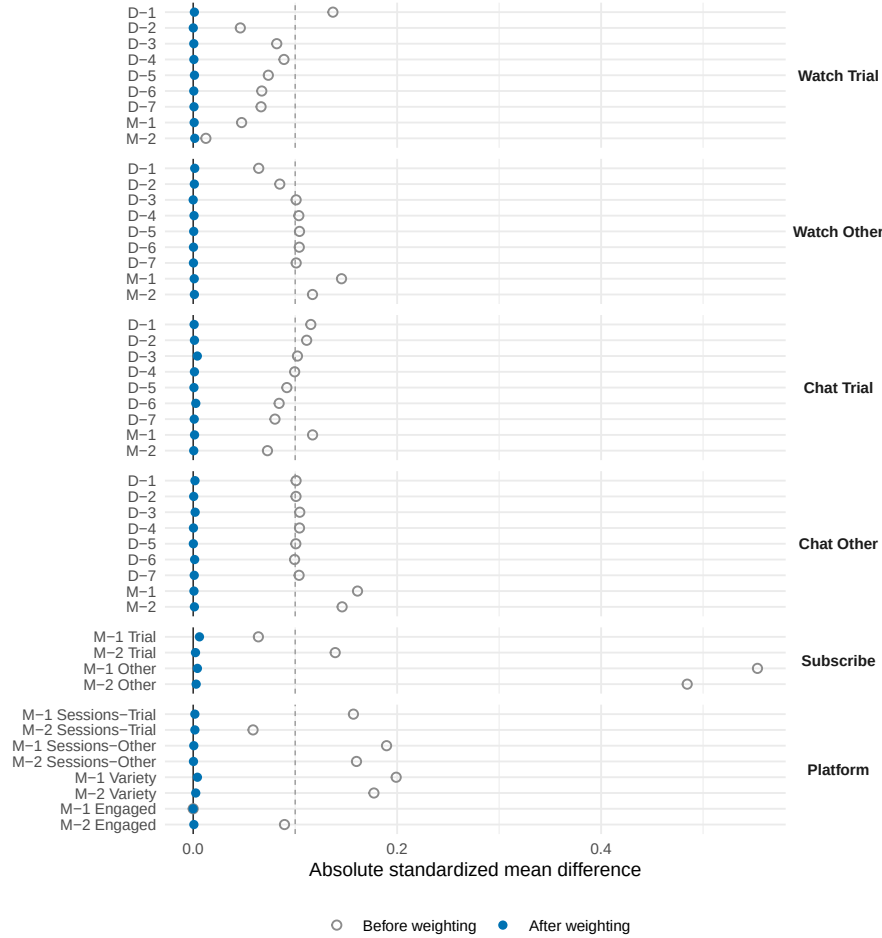
6.1 Estimated Assignment

Figure 2 shows absolute standardized mean differences in all pre-treatment covariates between treatment and control groups before and after overlap weighting. Before weighting, several covariates exceed the conventional 0.1 threshold for covariate balance (Austin, 2011). After applying the overlap weights, all differences are well below this threshold, indicating that treated and control users are balanced in terms of their observed characteristics. This diagnostic provides support for the conditional independence assumption. Additionally, although allocation depends on observed user behavior, our subsequent policy analysis finds

¹²A similar approach is used by Ellickson et al. (2023), who study the causal effects of targeted email promotions on purchase decisions, though that analysis benefited from observing the exact set of targeting variables used to assign treatment. In our case, we include as many potentially relevant covariates as possible, and rely on the method to identify the relative importance of features. For a deeper discussion of applications of double/debiased machine learning methods in the marketing literature, see Lemmens et al. (2026).

that observed assignment does not generate larger treatment effects than random allocation within the same cohorts ([Section 7](#)).

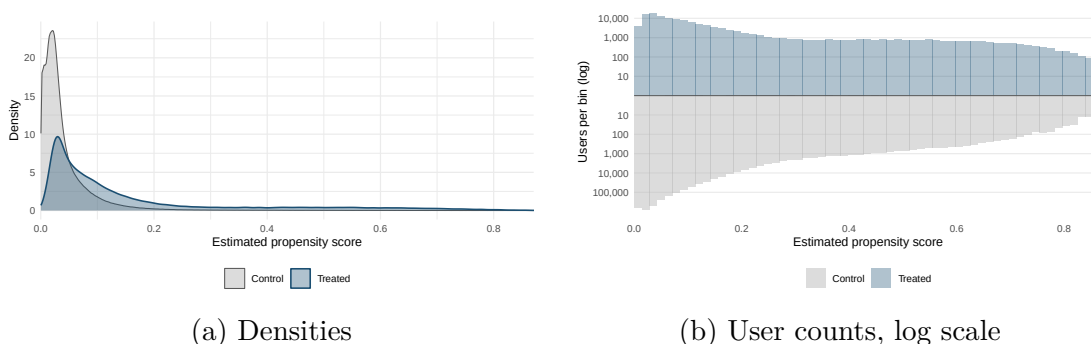
Figure 2: Covariate Balance



Notes: Absolute standardized mean differences in pre-treatment user behaviors between treatment and control groups, before weighting (open grey circles) and after overlap weighting (filled blue circles). Covariates are grouped into panels by behavior: watch, chat, subscriptions, and platform activity. The dashed line at 0.1 indicates the conventional balance threshold.

[Figure 3](#) shows the estimated propensity score (treatment-probability) distributions for treated and control users under the assignment model. The distributions overlap across the support, including in the tails, where the log-scale count view in panel (b) shows both groups populated in every bin. This diagnostic provides support for the overlap assumption.

Figure 3: Propensity Score Overlap



Notes: Estimated treatment-probability (propensity score) distributions for treated and control users from the assignment model. Panel (a) plots the propensity densities, each normalized within group. Panel (b) shows the same scores as two histograms, treated above and control below the x-axis. In both panels the x-axis is truncated at the 99.9th percentile of the treated distribution.

6.2 Average Treatment Effects

Table 2 presents average treatment effects for watch, chat, subscription, and platform outcomes across trial and other channels, and over the four months following the start of treatment. Each watch and chat effect should be interpreted as an average change in the level of the outcome, while each subscribe effect should be interpreted as a change in the probability of holding a paid subscription during the month.

Watch Watch time on the trial channel increases by 212 minutes during the trial month. These elevated watch levels persist throughout the month-long benefit period and continue at least until the third month after the benefit period concludes. Promotion recipients also increase their viewership of other channels by 258 minutes during the benefit period, indicating positive viewership spillovers across the platform. These positive spillover effects persist in each of the three months after the benefit period concludes. The increase in watch time on other channels exceeds the increase on the trial channel in every month, and the gap widens over time: trial-channel gains fall from 212 to 90 minutes between the benefit month

Table 2: Average Treatment Effects

Outcome	Channel	Months since treatment			
		M1	M2	M3	M4
Watch	Trial	211.62 (7.64)	156.58 (7.02)	132.24 (6.66)	90.34 (5.82)
	Other	258.31 (33.29)	275.45 (29.60)	269.39 (27.70)	219.11 (23.97)
Chat	Trial	9.65 (0.37)	3.24 (0.41)	2.94 (0.38)	2.76 (0.33)
	Other	3.45 (0.38)	3.89 (0.40)	3.63 (0.54)	2.74 (0.44)
Subscribe	Trial	–	-0.007 (0.001)	0.016 (0.001)	0.089 (0.001)
	Other	0.049 (0.001)	0.058 (0.001)	0.040 (0.001)	0.034 (0.001)
Sessions	Trial	2.751 (0.073)	2.084 (0.069)	1.843 (0.067)	1.326 (0.058)
	Other	4.126 (0.325)	4.619 (0.284)	4.346 (0.295)	3.365 (0.248)
Variety	All	0.249 (0.011)	0.302 (0.011)	0.302 (0.011)	0.261 (0.010)
Engaged	All	–	0.022 (0.001)	0.026 (0.001)	0.035 (0.001)

Notes: Each cell shows the overlap-weighted ATE from our DML estimator. The allocation model, $\hat{e}(x)$, is common across every cell; the pooled outcome models, $\hat{m}_j(x)$, are specific to each cell. Columns report months since the start of treatment. Cells marked – indicate effects not reported because the outcomes are mechanically affected by allocation or eligibility. Standard errors clustered at the cohort level are in parentheses. All effects are statistically significant at the $p < 0.001$ level.

and the third month after it expires, while other-channel gains remain above 200 minutes throughout.

These results show that promotion recipients exhibit significant and sustained increases in viewership across the platform. Taken together, the promotion has an expansionary effect on platform watch time rather than merely shifting time from other channels to the trial

channel. The persistent increases beyond the 30-day benefit period cannot be explained only by concurrent access to the functional benefits of a subscription. This suggests that temporary premium benefits influence some users’ content preferences and affinity towards the platform.

Chat Users who receive the promotion also significantly increase their use of the platform’s chat feature during and after the benefit period. Chat activity during the benefit period increases by 9.6 messages on the trial channel and 3.5 messages across all other channels. These increases are large, as fewer than 50% of users chat in the month prior to treatment on any of the observed channels. While the large initial increase on the trial channel is, in part, explained by promotion recipients writing “thanks” or “thank you” to the user who gifted the subscription, chat activity remains elevated across the trial and other channels for the three months after the benefit period concludes. These effects suggest that the promotion’s impact extends well beyond these initial reactions, indicating lasting and more engaged user behavior.

Paid Subscriptions Temporary premium benefits have large effects on subsequent subscription behavior that vary by channel and time horizon. Benefit recipients are initially 0.7 percentage points less likely to be subscribed to the trial channel in the month immediately following the benefit period, but this negative effect turns positive by the second month after the benefit period ends (1.6 percentage points) and reaches 8.9 percentage points by the third month. Conversely, subscription rates for other channels increase across all time horizons. Promotion recipients are 4.9 percentage points more likely to be subscribed to other channels during the benefit period. Treated users are also 5.8 percentage points more likely to subscribe to other channels in the month after the benefit period ends, with that pattern continuing for months three and four. These effects are substantial relative to the corresponding baselines. Because the estimand is an overlap-weighted average treatment effect, the most relevant and conservative comparison is the untreated mean in the overlap

population rather than in the full eligible sample. Overlap weighting concentrates on users whose allocation probabilities are away from zero and one, and those users subscribe more often than the average eligible user. Other-channel subscription rates among untreated users in the overlap population range from 6.9 to 10.1 percent over these months ([Table B.1](#)), so the estimated effects represent increases of roughly one half to three quarters over baseline. Despite the temporary reduction in propensity to subscribe to the trial channel, the net platform-wide effects on paid subscriptions overall, and during each of the three post-benefit months, are positive.

The initial reduction in trial channel subscriptions may reflect several behavioral mechanisms. Users may engage in intertemporal substitution, delaying purchases after receiving free access to the same benefits. Alternatively, content satiation may lead users to explore other creator content after intensive exposure to the trial channel. Another possible explanation is that receiving premium benefits delays the next opportunity to subscribe for treated users—they cannot purchase a new subscription until the start of month two at the earliest. In contrast, control users can purchase subscriptions throughout the benefit period. This creates a reduction in observed treatment group subscriptions in month two. The impact of this mechanism diminishes in months three and four as both groups have more equal opportunity to subscribe. In [Section 6.3](#), we explore where and for which users these subscription effects are most likely to occur.

Platform Outcomes Temporary premium benefits generate positive and lasting changes in platform-level engagement. During the benefit month, recipients watch, on average, 0.25 more channels (4.2% increase), tune in for 2.8 additional sessions on the trial channel (12.0% increase), and tune in for 4.1 additional sessions on the other channels (12.4% increase).¹³ The promotion increases the probability of remaining engaged—defined as any watch or chat activity on an observed channel during the month—by 2.2, 2.6, and 3.5 percentage points in

¹³Percentage increases throughout this section are computed against the overlap-weighted mean among untreated users, the untreated baseline for the population the overlap-weighted average treatment effect describes ([Table B.1](#)).

Months 2, 3, and 4, respectively. Against overlap-weighted baseline churn rates of 7.5, 12.0, and 20.8 percent, this corresponds to a decrease in churn of 30%, 21%, and 17% in the three months following the benefit period.

Overall, temporary premium benefits generate significant platform-wide expansionary effects. Users who receive benefits watch more, chat more, and are more likely to be subscribed in the long run to both the trial channel and other channels. These results highlight the importance of measuring and understanding the comprehensive effects of marketing interventions, as relevant effects may not be immediately evident and may extend beyond the immediate context of the treatment.

6.3 Understanding Cross-Channel Subscription Spillovers

The average treatment effects in [Table 2](#) show that temporary premium benefits generate significant subscription spillovers to other channels. In this subsection, we ask where and for which users those spillovers are most prevalent. The variation in our setting is not conducive to formally testing behavioral mechanisms, so we proceed descriptively. We state the predictions that candidate explanations make about where spillovers should concentrate, and we assess whether the estimated effects are consistent with those predictions. We first ask whether spillovers concentrate on other channels that are similar to the trial channel ([Sahni, 2016](#)). We then ask whether spillovers reflect users learning about the value of premium benefits ([Akerberg, 2001](#)).

Similarity We measure similarity in two ways. Audience similarity captures the overlap between the audiences of the trial channel and another channel in the 30 days before the allocation event, and content similarity captures the overlap in the content category labels

(e.g. game name) attached to each streaming session. We define similarity as pointwise mutual information between the two channels’ audiences and content, respectively.¹⁴

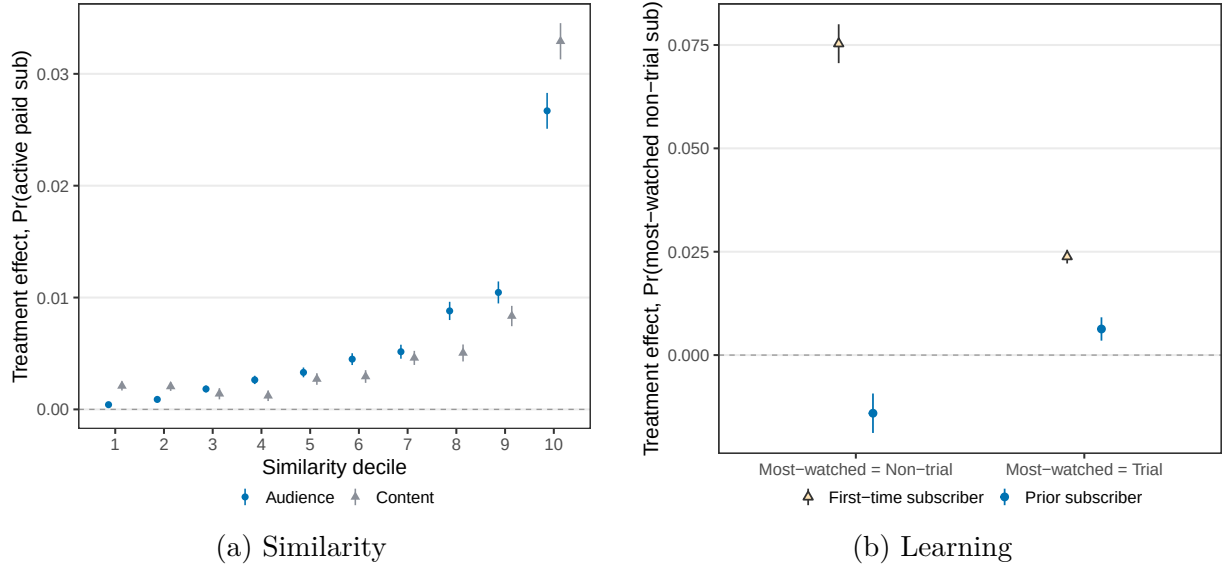
For each allocation event, we score every other observed channel’s similarity to that event’s trial channel and assign each event–channel pair to a decile of the resulting similarity distribution. We then define ten decile-specific outcomes. For each decile d , we define an indicator for holding a paid subscription during the benefit period to any other channel whose similarity to the trial channel falls in decile d . Panel (a) of [Figure 4](#) reports the overlap-weighted average treatment effect on each of these outcomes. For both the audience and content definitions of similarity, spillovers heavily concentrate on channels more similar to the trial channel, with effects increasing across deciles and reaching 53% and 67% of the total other-channel subscription effect at the top decile.¹⁵ Because channels similar to the trial channel are also those a user is more likely to watch already, this pattern is consistent both with increased engagement along a user’s existing preferences and with learning about the value of premium benefits on channels the user already watches. The similarity results alone cannot separate these explanations, so we next examine which users generate the spillovers and where they subscribe.

Learning We next ask whether the spillovers reflect users learning about the value of a subscription. Users without an observed prior paid subscription may have more to learn from a month of premium benefits than users with prior subscription experience ([Ackerberg, 2001](#)). If learning drives spillover subscriptions, two predictions follow. First, the response should be concentrated among users with no observed prior paid subscription, whom we refer to as *first-time subscribers*, while users with a prior subscription (*prior subscribers*) should have relatively smaller changes in behavior. Second, first-time subscribers should be more

¹⁴Pointwise mutual information, and the equivalent lift ratio, are standard measures of association between units of very different sizes ([Church and Hanks, 1990](#); [Netzer et al., 2012](#)). We provide additional details on the construction of these measures in [Online Appendix C](#).

¹⁵The decile-specific effects need not sum to the total Other subscription effect because users can subscribe to channels in more than one decile.

Figure 4: Variation in Cross-Channel Spillovers



Notes: Panel (a) shows overlap-weighted average treatment effects on paid subscriptions to other channels during the benefit period, where other channels are grouped into ten distinct outcomes by similarity to the trial channel. Blue circles refer to audience similarity and grey triangles to content similarity. Panel (b) shows best linear projections of overlap-weighted average treatment effects onto mutually exclusive subsets of users defined by subscription history (first-time versus prior subscriber) and by whether the trial channel is the user’s most-watched channel, where channels are ranked by watch hours over the 30 days before the allocation event. The outcome is an indicator for a paid subscription during the benefit period to the user’s most-watched other (non-trial) channel.

likely to subscribe where the benefits are most valuable to them. Because a primary benefit of a subscription is ad-free viewing on that channel, we argue that the value of a subscription to a channel is increasing in the time the user spends watching it. Therefore, new subscriptions should be directed toward channels the user spends more time watching. The alternative is that spillovers reflect prior subscribers reinforcing or expanding their existing subscription portfolios. In that case, the largest effects should appear among prior subscribers rather than first-time subscribers.

Panel (b) of Figure 4 examines these predictions. The outcome is an indicator for a paid subscription during the benefit period to the user’s most-watched other channel, where channels are ranked by watch hours over the 30 days before the allocation event. Because this outcome is undefined for users who watched only the trial channel, we restrict the sample to users who watched at least two observed channels in the 30-day interval leading up to the allocation event. We divide these users into four mutually exclusive groups defined by

subscription history (first-time versus prior subscriber) and by whether the trial channel is or is not the user’s most-watched channel. When the trial channel is not the user’s most-watched channel, the outcome refers to the channel the user watches most overall; when the trial channel is the most-watched channel, it refers to the user’s most-watched channel other than the trial channel. We fit a single causal forest on the restricted sample and report best linear projections of the overlap-weighted average treatment effect within each group.¹⁶

The results are consistent with learning. First-time subscribers respond by subscribing where premium benefits are more valuable to them: when their most-watched channel is not the trial channel, the promotion increases subscriptions to that channel by 7.5 percentage points from a baseline of 0.5%, and the effect is smaller when the trial channel is itself the user’s most-watched channel. Prior subscribers show muted effects. When their most-watched channel is not the trial channel, the promotion reduces subscriptions to that channel by 1.5 percentage points from a baseline of 47.1%, with a smaller effect when the trial channel is their most-watched channel. The large response among first-time subscribers, directed at their most-watched channel, matches both predictions of learning, and the muted and partly negative response among prior subscribers is inconsistent with the alternative that spillovers come from users expanding existing subscription portfolios.

Taken together, the evidence points to two forces working in opposite directions. The promotion creates new subscription demand among first-time subscribers, concentrated on channels that are similar to the trial channel and that the user already watches, while displacing some existing demand among prior subscribers on the channels they watch most. The largest response occurs when the promotion reaches first-time subscribers on a channel other than the one they watch most, so that the benefits are experienced outside the user’s preferred content.

¹⁶A best linear projection uses a linear model to summarize how treatment effects vary across user characteristics (Semenova and Chernozhukov, 2021). Here, those characteristics indicate membership in the four groups, so the model yields a debiased average treatment effect for each group.

6.4 Robustness and Sensitivity

Because our main identifying assumption of unconfoundedness cannot be directly tested, we investigate the robustness of the effects to alternative modeling and sampling decisions. Re-estimating every effect under alternative outcome definitions—inverse hyperbolic sine and log transformations, and binary thresholds across a range of cutoffs—results in the same sign and significance of each result ([Online Appendix D.1](#)). We also estimate effects consistent with our main results using an eligibility-discontinuity sample that compares users based on their arrival time into the channel, comparing eligible users who joined just before the allocation to otherwise similar users who joined just after and were therefore ineligible ([Online Appendix D.2](#)). The results are also consistent when we replace our preferred estimator with a variety of linear and machine-learning estimators ([Online Appendix D.3](#)).

Finally, we assess sensitivity to unobserved confounding by benchmarking the predictive strength needed to eliminate each effect against that of observed covariate groups ([Online Appendix D.4](#)). The Other subscription effects are particularly robust: an unobserved confounder would need 8.2–12.0 times the predictive strength of the most informative group of covariates in both the assignment and outcome models to reduce these estimates to zero.

Given the robustness of our results to this series of checks, we interpret our findings as causal effects of temporary premium benefits on the behavior of eligible users in the overlap population.

7 Multi-Objective Targeting

The preceding results show that temporary premium benefits affect a variety of user behaviors, including those beyond the promoted channel, and that these effects vary across users. In this section, we consider how a platform can use estimated effects to evaluate promotional targeting policies designed to optimize different, potentially competing, objectives. Specifically, we address three questions: i) How much can optimal policies improve outcomes

relative to random targeting? ii) What are the opportunity costs of designing a policy to optimize a single outcome? and iii) Can the platform design policies to optimize multiple outcomes while achieving comparable performance to single-objective policies? To do this, we use the multi-objective targeting framework of [Rafieian et al. \(2025\)](#).

7.1 Problem Definition

We consider a platform’s decision of how to allocate temporary premium benefits across a set of eligible users. We assume the platform values three known outcomes, though in practice the platform may care about any number of outcomes. We focus on paid subscriptions to the trial channel, paid subscriptions to other channels, and whether users stay engaged, all measured in Month 4. These outcomes capture monetization within the promoted channel, spillovers beyond it, and continued use of the platform. The treatment effects for these three outcomes are also among the most robust of our estimates. We use *targeted* to refer to users selected by a hypothetical policy to distinguish them from *treated* users who actually received benefits.

A policy π maps users’ characteristics X to an allocation decision, where $\pi(X) = 1$ indicates that the user is targeted to receive benefits. A policy’s performance on outcome j , $\rho_j(\pi)$, is the average of conditional treatment effects, $\tau_j(X)$, among the users it targets. The platform’s objective is a weighted sum of performance across the three outcomes. For a given set of objective weights, the platform solves

$$\max_{\pi \in \Pi(Z)} \sum_{j=1}^3 \beta_j \rho_j(\pi), \quad \rho_j(\pi) = \mathbb{E}[\tau_j(X) \mid \pi(X) = 1],$$

where $\Pi(Z)$ restricts each policy to targeting exactly Z_c users in cohort c , and the nonnegative weights β_j sum to one. For this exercise, we fix Z_c at five times the observed count of treated

users in cohort c .¹⁷ This larger targeting set provides a conservative test of the benefits to targeting because gains are easier to achieve when a policy can concentrate on a small number of users with the largest effects. In this setup, the weights represent the relative value the platform places on a percentage-point increase of each outcome. The optimal policy for a given set of objective weights targets the Z_c users in a cohort with the largest weighted sum of conditional treatment effects.

We explore three single-objective policies, each placing all weight on one outcome; an equal-weight policy, assigning one-third to each outcome; and three “anchored” policies that assign 60% weight to a focal outcome and divide the remaining 40% between the other two outcomes to maximize the sum of their predicted policy values.

7.2 Estimation and Evaluation

Implementing these policies requires estimates of conditional treatment effects to determine which users to target. However, evaluating these policies using the same estimates that determine whom they target can overstate their performance (Athey and Wager, 2021; Hitsch et al., 2024; Xu et al., 2025). We therefore evaluate policies using observed outcomes on a held-out sample rather than the predicted treatment effects used to construct them.

We divide the full set of cohorts into estimation, evaluation, and policy samples. The estimation sample is used to fit causal forests to estimate conditional treatment effects for the three outcomes.

We evaluate policy performance using an inverse-propensity-weighted (IPW) estimator. The evaluation sample is used to fit an assignment model that supplies predicted assignment probabilities $\hat{e}(X_i)$ to be used in the IPW estimator. While the causal forest’s assignment

¹⁷In our empirical setting, the number of allocated subscriptions is chosen by the users who purchase them rather than by the platform, and a change in the allocation rule could affect how many subscriptions are purchased. We interpret each policy as a reallocation of a fixed number of subscriptions within a cohort, but our estimates do not capture potential responses in the quantity purchased. Alternatively, the counterfactual policies can be interpreted as platform-initiated promotions, provided that the estimated responses to user-initiated gifts can be extrapolated.

model could also provide these estimates, reusing these estimates for the evaluation step could propagate estimation errors into both the targeting predictions and the policy evaluation.¹⁸

The policy sample consists of the remaining cohorts not used to fit the conditional treatment effect or assignment models. Using predictions from the causal forests fitted on the estimation sample, each policy ranks users within policy sample cohorts by the weighted sum of their conditional treatment effects and targets the Z_c highest-ranked users.¹⁹ Let $T(\pi)$ denote the set of policy sample users targeted by policy π . Using observed outcomes and predicted assignment probabilities in the policy sample, we estimate policy performance with the following normalized IPW estimator:

$$\hat{\rho}_j(\pi) = \frac{\sum_{i \in T(\pi)} W_i Y_{ij} / \hat{e}(X_i)}{\sum_{i \in T(\pi)} W_i / \hat{e}(X_i)} - \frac{\sum_{i \in T(\pi)} (1 - W_i) Y_{ij} / (1 - \hat{e}(X_i))}{\sum_{i \in T(\pi)} (1 - W_i) / (1 - \hat{e}(X_i))}. \quad (5)$$

[Online Appendix E](#) provides additional estimation details and examines the robustness of the targeting results to alternative conditional treatment effect and policy evaluation estimators.

7.3 Policy Performance

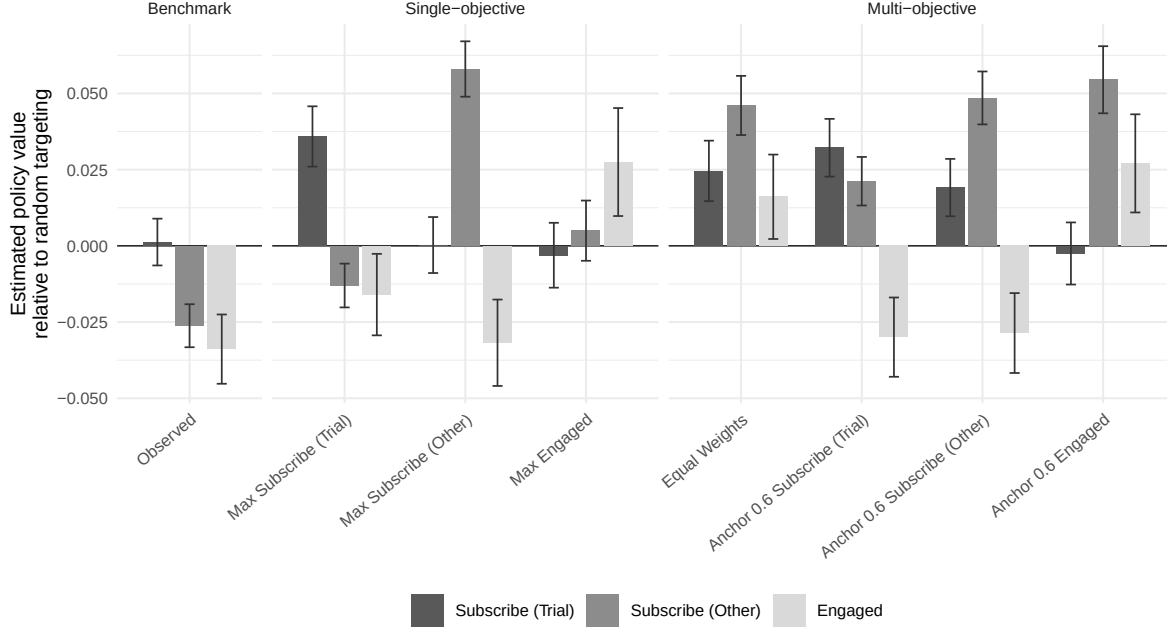
[Figure 5](#) reports differences in policy performance relative to within-cohort random targeting. The platform’s observed allocation produces smaller effects than random targeting on Other subscriptions and Engaged, by 2.6 and 3.4 percentage points, respectively, and is not statistically different from random targeting for Trial subscriptions. These comparisons do not provide strong evidence that the platform’s observed allocation rule targets users to optimize any of these outcomes or any other outcomes strongly correlated with these outcomes.

Each single-objective policy produces a significantly larger effect on its focal outcome: 3.6 percentage points for Trial subscriptions, 5.8 points for Other subscriptions, and 2.7 points

¹⁸For IPW evaluation, we truncate predicted assignment probabilities to $[0.01, 0.99]$ to limit extreme propensity weights.

¹⁹The same predictions determine how the anchored policies divide the remaining 40% weight between the two non-focal outcomes.

Figure 5: Policy Performance Relative to Random Targeting



Notes: Differences in estimated policy values relative to within-cohort random targeting. Observed allocation is evaluated using the average treatment effect on actual recipients. Error bars reflect 95% confidence intervals for paired contrasts, clustered by cohort.

for Engaged. However, these gains can come at the expense of effects on other outcomes. Relative to the Equal Weights policy, the policy optimized solely for Trial subscriptions increases that effect by 1.1 percentage points but reduces the Other subscription effect by 5.9 points and the Engaged effect by 3.2 points. The policy optimized solely for Other subscriptions reduces the Engaged effect below random targeting, by 3.2 points.

Multi-objective policies allow the platform to balance these gains and losses. The Equal Weights policy is the only policy we consider that significantly improves all three outcomes relative to random targeting, by 2.5, 4.6, and 1.6 percentage points, respectively. Compared with the policy optimized solely for Engaged, the policy assigning 60% weight to Engaged increases the Other subscription effect by 5.0 percentage points while leaving the Trial subscription and Engaged effects essentially unchanged. Whether this tradeoff is desirable depends on the platform's relative valuation of the outcomes.

These results show that pursuing the largest possible effect on one outcome can come at a substantial cost to others. A policy that accepts a small reduction in one effect can achieve much larger gains elsewhere.

8 Conclusion

In this paper, we study the effects of temporary premium benefits on user behavior. Our analysis centers on a multi-channel social livestreaming platform where users can purchase premium subscriptions to specific creator channels. We leverage quasi-exogenous variation in the promotional allocation of premium benefits together with a flexible double/debiased machine learning approach to measure causal effects across multiple outcomes and time horizons. The promotion we study not only affects future subscription purchases, but also a wide variety of platform-relevant behaviors, including watch activity and social engagement. These effects vary meaningfully across time and for different user segments, and spill over to other channels on the platform. Users who receive benefits on a channel they watch less are more likely to respond by subscribing to other channels that they watch more regularly. We quantify the tradeoffs in designing targeting policies to optimize different objectives and show that single-outcome targeting carries high opportunity costs due to the potential for promotions to influence a broad set of platform-relevant outcomes.

Our results have several implications for platforms and marketing managers. First, firms benefit from taking a holistic view when evaluating marketing interventions. A narrow focus on individual metrics or short-term outcome windows may misrepresent time-varying treatment effects and obscure tradeoffs between objectives. Second, the heterogeneity results suggest that targeting users outside their most-watched channel may help platforms generate cross-channel spillovers. This result applies specifically to firms that sell multiple products or services. Although free access to premium benefits may cannibalize purchases in the short term, multi-service firms can offset these negative effects by internalizing positive spillovers

across their portfolio. For example, a single-product firm (e.g., Netflix) cannot capture cross-service spillovers from promotions like free trials. In contrast, a multi-product firm (e.g., Disney, which offers multiple streaming services and other entertainment products) stands to benefit when a promotion on one service increases purchases or engagement on another. Importantly, a multi-service firm must analyze user data across its entire portfolio, as evaluating each service in isolation would miss effects that determine overall profitability.

Our analysis leaves several questions for future research. The spillover patterns across channel similarity, subscription history, and viewing preferences are consistent with learning about subscription benefits, but do not distinguish this explanation from substitution, complementarity, or channel quality differences. Experiments varying these features could clarify the mechanisms and inform promotion design. Our reduced-form estimates also do not identify the underlying preferences linking time allocation to willingness to pay. Finally, implementing effective targeting requires firms to balance experimentation, which supports learning about preferences and treatment effects, with exploitation of current knowledge to improve short-term performance. Managing this trade-off as user preferences and content offerings evolve remains an important operational challenge.

Declarations

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Online Appendix

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A Data and Sample

A.1 Creator Sample

Twitch has over 1 million content creators (streamers), each with their own channel; however, many of those streamers have few or no viewers. For tractability, we focus the analysis on the top 100 English language streamers. We used a Twitch data aggregator¹ to select the 100 highest-ranked English language streamers by total viewership in the 90 days leading up to the start of the sample (July 1, 2022), and who are also ranked in the top 150 by watch time in the 30 days leading up to the start of the sample. The 90-day selection criterion helps define streamers with a sustained presence on the platform, while the 30-day criterion helps identify streamers who were currently active and popular on the platform. Table A.1 contains the full list of 100 streamers included in our sample. While the majority of these channels are operated by individuals, a select few (e.g. `esl_csgo`) are managed by organizations. These organization-managed channels may not have many active subscribers, but they are useful for measuring spillovers in watch and engagement.

Table A.1: Creator Channels in Sample

39daph	daltoosh	ironmouse	ninja	smitegame
aceu	deuceace	jinnytty	nmplol	sodapoppin
adinross	disguisedtoast	k3soju	northernlion	summit1g
amouranth	distortion2	kyedae	otzdarva	swagg
asmongold	doublelift	kyle	penta	sweetdreams
aydan	dreamleague	lck	pestily	symfuhny
barbarousking	elajjaz	lcs	pokelawls	sypherpk
beyondthesummit	esfandtv	lec	pokimane	tarik
blastpremier	esl_csgo	leekbeats	prod	tenz
boxbox	esl_dota2	lirik	quickybaby	tfue
brawlhalla	fextralife	loltyler1	quin69	thebausffs
brucedropemoff	foolish_gamers	lpl	rainbow6	topsonous
buddha	forsen	lvndmark	ranboolive	trainwreckstv
castro_1021	fuslie	mande	rocketleague	tsm_imperialhal
chess	gamesdonequick	maximilian_dood	roshtein	uberhaxornova
chilledchaos	gorgc	mizkif	saintone	valorant
classybeef	gtawiseguy	moister1tikal	shahzam	xqc
clix	hasanabi	moonmoon	shivfps	yourragegaming
coconutb	hiko	nickeh30	shroud	zealsambitions
cohhcarnage	iitztimmy	nickmercs	sinatraa	zerkaa

¹<https://sullygnome.com>

A.2 Data Collection

Viewing Behavior

We collected watch data by querying the <https://tmi.twitch.tv/> API endpoint. This endpoint returned the broadcaster’s name, live status, and a list of logged-in users connected to the channel’s chat, which we use to measure channel presence and infer watch time. We rotated through 100 streamers and queried the endpoint for each streamer approximately once per five minutes from July 2022 until the endpoint was officially shut down in early April 2023.² The approximately five-minute frequency of collection ensures at least one observation per channel in the six-minute window leading up to any cohort start time, which we use in our definition of eligible users.

Short data gaps occurred due to power outages and reaching device storage limits. [Figure A.1](#) shows the distribution of successful API calls over the sample period. The maximum height of the bars is determined by the cadence of our data collection procedure. We had four separate computers running, each collecting 1/4 of the data. The first dip (09/2022) affected all computers; the subsequent dips each only affected one computer. The dips resulted from human/technology/utilities-related failures in our data collection infrastructure rather than Twitch service outages that could be correlated with treatment. We therefore have no reason to believe these short outages will affect the results in any substantive way.

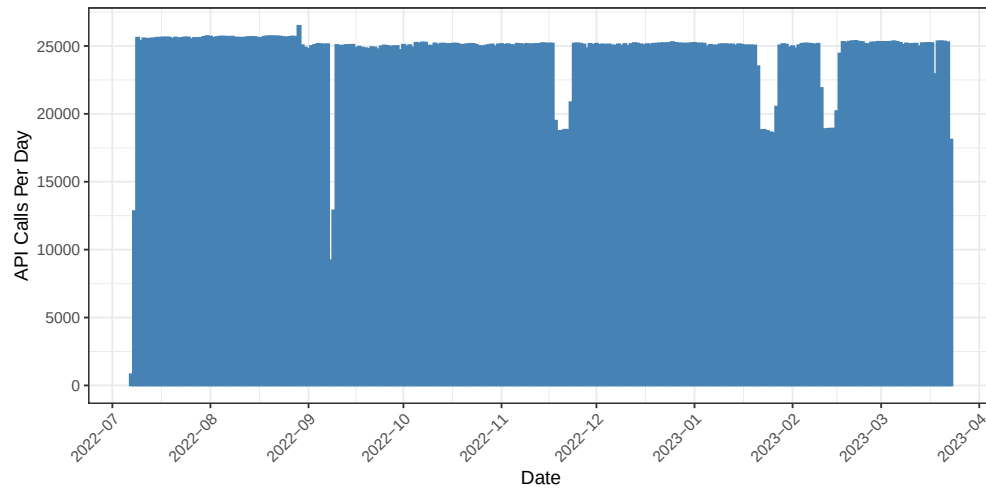
Chat and Subscription Behavior

Channel-specific chat logs were collected using the Chatty application,³ which allowed for the simultaneous monitoring of all channels in our sample. For each channel, we recorded the complete chat transcript, including all messages sent and subscription status change notifications. These data were collected on a different computer than the viewing behavior.

²See <https://discuss.dev.twitch.com/t/legacy-chatters-endpoint-shutdown-details-and-timeline-april-2023/43161> for the official announcement.

³<https://chatty.github.io/>

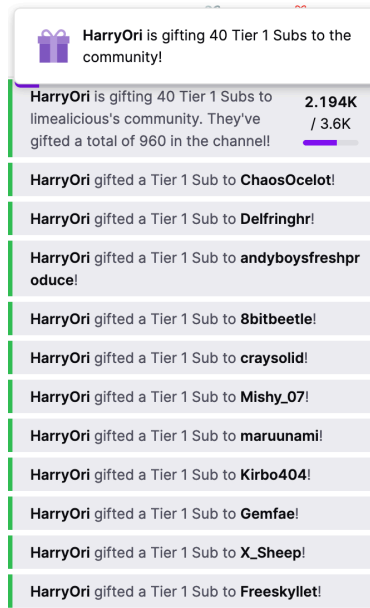
Figure A.1: Viewing Behavior Collection Outages



Notes: Successful viewing-data API calls per day.

Figure A.2 shows an example of a channel’s chat feed where a user purchased 40 allocated subscriptions for other eligible users on the channel.

Figure A.2: Algorithmically-Allocated Subscription Announcement



Notes: Chat announcement of a purchase of 40 algorithmically allocated subscriptions.

A.3 Data Cleaning and Processing

Our estimation sample is comprised of cohorts with treatment dates ranging from September 2022 to December 2022 to ensure each cohort includes a two-month pre-treatment and four-month post-treatment observation period for all eligible users.

We limit our analysis to cohorts with at least five treated users. The primary motivation for this filter is computational ease, as each cohort used in the estimation requires computing cohort-specific covariates and outcomes for a separate set of control users. Among algorithmically-allocated events in our panel window (those with two or more recipients) allocation events with five or more recipients account for 95% of all recipients. The average number of treated users per cohort in the final estimation sample is 10.4. We further restrict the sample to eligible users with at least 30 minutes of total watch time on the trial channel, and no more than 21,600 minutes (twelve hours per day) of total watch time across all observed channels, during the month before the allocation. The floor filter removes users with only incidental exposure to the trial channel; the cap excludes users whose watch levels are more consistent with bot activity than with human viewing. Among the 142,714 eligible treated users who were not already subscribed to the trial channel, the viewership floor removes 2,556 (1.79%). The viewership cap removes an additional 3,734 (2.62%), leaving 136,424 treated users.

A.4 Descriptive Statistics

[Table A.2](#) provides summary statistics for every pre-treatment behavior, pooled over treated and control users.

[Table A.3](#) presents pre-treatment behavior descriptives separately for the treated and control users. There are differences in baselines between the two groups, as shown in balance tests reported in [Figure 2](#). Treated users watch the trial channel less during the 24 hours before allocation (130.52 versus 155.23 minutes), but post more chat messages (2.30 versus

Table A.2: Pre-Treatment Covariates

Behavior	Channel	Days Pre							Months Pre	
		D-1	D-2	D-3	D-4	D-5	D-6	D-7	M-2	M-1
Watch	Trial	154.05 (196.98)	79.56 (164.62)	70.44 (151.64)	64.96 (144.67)	63.02 (139.88)	61.31 (136.44)	61.40 (133.35)	1,335.07 (2,343.77)	1,708.91 (2,362.03)
	Other	70.05 (192.06)	67.57 (167.16)	65.36 (161.21)	63.34 (156.15)	63.09 (154.76)	63.64 (154.14)	64.45 (154.70)	1,890.57 (3,559.58)	1,878.55 (3,070.09)
Chat	Trial	1.10 (10.06)	0.56 (7.63)	0.52 (7.34)	0.46 (6.86)	0.45 (6.73)	0.43 (6.63)	0.44 (6.62)	10.58 (117.20)	13.06 (127.13)
	Other	0.37 (6.37)	0.35 (6.06)	0.33 (5.52)	0.31 (5.11)	0.31 (5.28)	0.31 (5.44)	0.32 (5.31)	8.72 (92.84)	9.41 (96.91)
Subscribe	Trial	—	—	—	—	—	—	—	0.022 (0.147)	0.016 (0.125)
	Other	—	—	—	—	—	—	—	0.023 (0.150)	0.026 (0.159)
Sessions	Trial	—	—	—	—	—	—	—	16.32 (23.94)	21.08 (24.55)
	Other	—	—	—	—	—	—	—	31.13 (51.06)	31.40 (46.53)
Variety	All	—	—	—	—	—	—	—	5.31 (4.94)	5.76 (4.76)
Engaged	All	—	—	—	—	—	—	—	0.902 (0.298)	1.000 (0.000)

Notes: Means, with standard deviations in parentheses, pooled across treated and control users. D-1–D-7 are daily pre-treatment windows; M-2 and M-1 are the preceding 30-day windows. Engaged equals 1 in M-1 by construction, because eligibility requires the user to be watching the trial channel at the time of allocation. Cells marked — are not measured. The sample contains 2,862,563 user-cohort observations across 13,089 cohorts.

1.04). Treated users also show higher subscription rates to other channels at baseline (0.176 versus 0.018 in M-1; 0.145 versus 0.017 in M-2) and greater engagement with the broader platform. These differences highlight the importance of accounting for pre-treatment covariates.

Table A.3: Covariate Descriptives by Treatment

<i>Treated Users</i>										
Behavior	Channel	Days Pre							Months Pre	
		D-1	D-2	D-3	D-4	D-5	D-6	D-7	M-2	M-1
Watch	Trial	130.52 (160.09)	86.43 (146.31)	81.91 (141.55)	77.05 (140.16)	72.64 (133.86)	69.92 (132.15)	69.75 (129.81)	1,309.05 (2,010.06)	1,807.99 (1,983.54)
	Other	81.18 (170.02)	81.16 (169.13)	81.16 (167.62)	79.24 (166.65)	79.00 (166.17)	79.51 (166.55)	79.87 (166.40)	2,296.16 (3,728.42)	2,321.08 (3,328.16)
Chat	Trial	2.30 (11.71)	1.46 (9.41)	1.40 (10.52)	1.16 (7.87)	1.10 (8.16)	0.99 (7.35)	1.00 (8.00)	18.82 (120.49)	27.62 (134.36)
	Other	1.25 (11.56)	1.21 (11.41)	1.17 (10.85)	1.11 (10.39)	1.10 (10.45)	1.06 (9.83)	1.11 (10.22)	29.94 (198.83)	32.51 (193.27)
Subscribe	Trial	–	–	–	–	–	–	–	0.046 (0.209)	0.024 (0.154)
	Other	–	–	–	–	–	–	–	0.145 (0.353)	0.176 (0.381)
Sessions	Trial	–	–	–	–	–	–	–	17.60 (21.48)	24.48 (20.66)
	Other	–	–	–	–	–	–	–	39.36 (57.07)	40.35 (52.70)
Variety	All	–	–	–	–	–	–	–	6.17 (5.35)	6.69 (5.14)
Engaged	All	–	–	–	–	–	–	–	0.926 (0.262)	1.000 (0.000)
<i>Control Users</i>										
Behavior	Channel	Days Pre							Months Pre	
		D-1	D-2	D-3	D-4	D-5	D-6	D-7	M-2	M-1
Watch	Trial	155.23 (198.57)	79.21 (165.47)	69.87 (152.11)	64.35 (144.87)	62.54 (140.15)	60.88 (136.63)	60.99 (133.51)	1,336.37 (2,359.22)	1,703.95 (2,379.28)
	Other	69.49 (193.08)	66.89 (167.03)	64.57 (160.85)	62.54 (155.56)	62.29 (154.12)	62.85 (153.45)	63.67 (154.05)	1,870.28 (3,549.70)	1,856.41 (3,054.92)
Chat	Trial	1.04 (9.97)	0.52 (7.53)	0.48 (7.14)	0.42 (6.80)	0.42 (6.64)	0.40 (6.59)	0.41 (6.54)	10.17 (117.01)	12.33 (126.71)
	Other	0.32 (5.99)	0.31 (5.66)	0.29 (5.11)	0.27 (4.68)	0.27 (4.87)	0.28 (5.12)	0.28 (4.93)	7.65 (83.95)	8.25 (89.24)
Subscribe	Trial	–	–	–	–	–	–	–	0.021 (0.143)	0.015 (0.123)
	Other	–	–	–	–	–	–	–	0.017 (0.129)	0.018 (0.134)
Sessions	Trial	–	–	–	–	–	–	–	16.26 (24.05)	20.91 (24.71)
	Other	–	–	–	–	–	–	–	30.72 (50.70)	30.95 (46.16)
Variety	All	–	–	–	–	–	–	–	5.26 (4.91)	5.71 (4.74)
Engaged	All	–	–	–	–	–	–	–	0.900 (0.299)	1.000 (0.000)

Notes: Means, with standard deviations in parentheses, reported separately for treated and control users. D-1–D-7 are daily pre-treatment intervals; M-2 and M-1 are the preceding 30-day intervals. Engaged equals 1 in M-1 by construction, because eligibility requires the user to be watching the trial channel at the time of allocation. Cells marked – are not measured. The sample contains 136,424 treated and 2,726,139 control user-cohort observations across 13,089 cohorts.

B Estimation

B.1 Hyperparameter Tuning

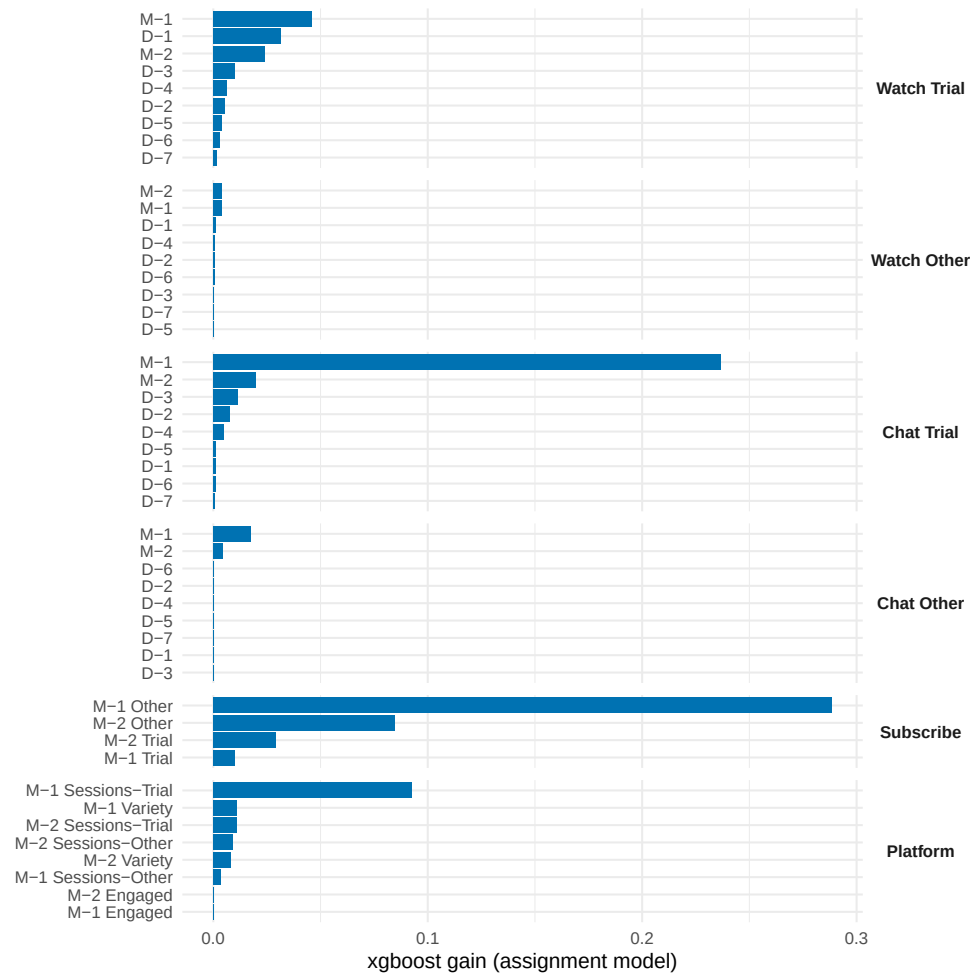
We tune the hyperparameters of the two gradient-boosted nuisance models: the assignment model $e(x)$ and the pooled outcome model $m_j(x)$. The assignment model is scored by out-of-fold log loss. Because one set of outcome-model settings is shared across every outcome, the outcome model is scored by out-of-fold R^2 averaged over three representative outcomes measured during the benefit period: one continuous outcome (trial-channel watch time) and two binary outcomes (chat activity on the trial channel and paid subscriptions to other channels). Tuning runs on a 20% sample of cohorts with 5-fold cross-validation, assigning whole cohorts to folds so that no cohort spans a fold. The five folds are used for hyperparameter tuning; nuisance predictions used in estimation are cross-fitted over two cohort-level folds. We search six hyperparameters sequentially over a coarse grid: the number of rounds (nrounds), learning rate (eta), maximum tree depth (max_depth) and minimum child weight (min_child_weight) first and then, holding those at their selected values, the row and column subsampling fractions (subsample and colsample_bytree). All other boosting parameters, including the minimum loss reduction (gamma) and the loss weight on treated units (scale_pos_weight), are left at their defaults. The assignment model uses nrounds = 600, eta = 0.05, max_depth = 4, min_child_weight = 20, subsample = 0.8 and colsample_bytree = 0.8; the outcome models use nrounds = 600, eta = 0.05, max_depth = 8, min_child_weight = 1, subsample = 0.8 and colsample_bytree = 0.6. Predictive performance varies little across a broad range of the tested hyperparameter settings for both the assignment and outcome models.

B.2 Variable Importance

Figure B.1 shows variable importance measures from the assignment model. Chat activity on the trial channel and subscriptions to other channels during the first pre-treatment

month (M-1) are the strongest predictors of assignment. Session counts and watch activity on the trial channel also strongly predict assignment, as do subscribe-related covariates. In contrast, watch time and chat activity on other channels are not strong predictors of treatment assignment. This pattern demonstrates that user behavior on the trial channel is a key determinant of assignment, while activity elsewhere on the platform has less influence.

Figure B.1: Assignment Model Variable Importance



Notes: Gain measures the contribution of each assignment model covariate to predicting allocation.

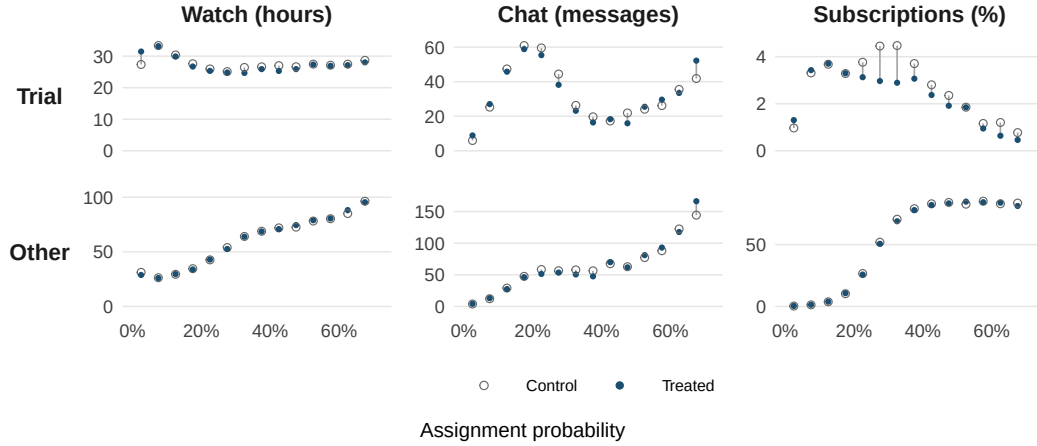
B.3 Pre-Treatment Behavior by Assignment Probability

Before weighting, treated users have higher subscription rates to other channels than control users ([Table A.3](#)). To examine this difference, we compare pre-treatment behavior among users with similar predicted assignment probabilities. [Figure B.2](#) shows mean behavior in M-1 within five-percentage-point intervals of predicted assignment probabilities. Subscription rates to other channels increase with the predicted assignment probability but are similar between treated and control users within these intervals. Watch time and chat activity also generally follow similar patterns across the two groups. There are small differences in subscription rates to the trial channel. One possible explanation is that we observe a past trial-channel subscription but miss an active annual or Prime subscription at allocation. Such users would remain in our control sample even though their active subscription makes them ineligible to receive benefits. If this occurs more often among users with recorded M-1 trial-channel subscriptions, it would raise the observed subscription rate among controls relative to treated users and could account for part of the gap. Together with the distributions in [Figure 3](#), these patterns suggest that the raw differences in other-channel subscription rates largely reflect the different distributions of treated and control users across predicted assignment probabilities, rather than large differences among users with similar predicted assignment probabilities.

B.4 Overlap-Weighted Baselines

[Table B.1](#) presents the overlap-weighted mean and standard deviation of each outcome among control users. Each cell estimates the mean untreated outcome for the population described by the overlap-weighted average treatment effects in [Table 2](#). This is the appropriate and conservative denominator for expressing overlap-weighted treatment effects in percentage terms. The corresponding unweighted means over the eligible sample are reported in [Table 1](#) and are substantially smaller, because overlap weighting concentrates on users whose allocation

Figure B.2: Pre-Treatment Behavior by Predicted Assignment Probability



Notes: Unweighted means of pre-treatment user behaviors in M-1, by five-percentage-point intervals of out-of-sample predicted assignment probabilities. Subscriptions denotes the percentage holding a paid subscription. The x-axis is truncated at the pooled 99.9th percentile. Vertical scales vary across panels.

probabilities are away from zero and one and who are more active and more subscription-prone than the average eligible user. Row and column definitions follow [Table 1](#).

Table B.1: Overlap-Weighted Baselines

Behavior	Channel	Months Pre		Months Post			
		M-2	M-1	M1	M2	M3	M4
Watch	Trial	1,342.36 (2,063.73)	1,833.25 (2,046.68)	1,864.09 (2,339.80)	1,282.64 (2,099.45)	1,119.92 (2,007.11)	816.59 (1,755.43)
	Other	2,021.69 (3,504.91)	2,050.93 (3,101.64)	2,033.65 (3,630.47)	1,789.51 (3,572.71)	1,676.95 (3,612.61)	1,270.57 (3,089.61)
Chat	Trial	18.15 (124.56)	25.71 (135.42)	25.33 (147.11)	18.18 (135.34)	16.32 (136.24)	14.93 (131.12)
	Other	21.22 (153.40)	23.21 (159.22)	21.07 (160.34)	18.58 (159.62)	17.70 (174.54)	16.51 (159.94)
Subscribe	Trial	0.043 (0.204)	0.025 (0.158)	0.039 (0.193)	0.045 (0.206)	0.042 (0.200)	0.039 (0.194)
	Other	0.098 (0.297)	0.112 (0.315)	0.101 (0.301)	0.080 (0.271)	0.073 (0.261)	0.069 (0.253)
Sessions	Trial	17.70 (21.65)	24.43 (21.22)	22.91 (23.52)	16.08 (21.08)	14.11 (20.35)	10.37 (18.26)
	Other	34.33 (52.14)	35.41 (47.84)	33.36 (50.26)	28.14 (47.26)	26.03 (47.92)	19.85 (41.58)
Variety	All	5.71 (5.03)	6.23 (4.85)	5.87 (4.63)	5.04 (4.47)	4.64 (4.41)	3.69 (4.19)
Engaged	All	0.921 (0.270)	1.000 (0.000)	0.999 (0.036)	0.925 (0.264)	0.880 (0.325)	0.792 (0.406)

Notes: Propensity-weighted means among untreated users, with weighted standard deviations in parentheses. Time intervals follow [Table 1](#). The Subscribe-Trial M1 cell (0.039) is the overlap-weighted rate at which *control* users subscribed to the trial channel during the benefit month, and differs from the pooled mean of 0.067 in [Table 1](#), which includes the treated users' allocated subscriptions.

C Audience and Content Similarity

This section describes the two similarity measures used in [Figure 4](#). We define each measure, describe the data used to construct it, and provide a comparison of the two measures.

C.1 Definitions

Audience Similarity. Let V_{cb} denote the set of distinct users that watch channel c during the 30 days preceding the allocation event b . For a trial channel c and other channel j , we define audience similarity as the pointwise mutual information (PMI) between the two viewing events,

$$\text{PMI}_{cjb} = \frac{\Pr(c, j)}{\Pr(c) \Pr(j)} = \frac{N |V_{cb} \cap V_{jb}|}{|V_{cb}| |V_{jb}|}, \quad (6)$$

where N is the number of distinct users on the platform. [Equation 6](#) normalizes by the product of the two audience sizes rather than by their union, so it measures overlap relative to what the two channels’ sizes alone would predict and equals one when viewing the two channels is independent. PMI is a useful measure because it facilitates comparisons across audiences of different sizes. Alternatives (e.g., the Jaccard index) that measure similarity as the share of viewers among a combined audience (or intersection) are more influenced by the audience size of individual channels ([Church and Hanks, 1990](#); [Netzer et al., 2012](#)).

Content Similarity. Let h_{ckt} be the hours channel c streamed content category k (e.g. a game name or “Just Chatting”) in month t . We measure content by month rather than by cohort to simplify data collection, as streamers’ content choices are highly persistent across time. The channel’s content profile is its within-channel share,

$$p_{ckt} = \frac{h_{ckt}}{\sum_{k'} h_{ck't}}, \quad \sum_k p_{ckt} = 1. \quad (7)$$

Feature popularity in month t is the corresponding share of all streamed hours across channels,

$$q_{kt} = \frac{\sum_c h_{ckt}}{\sum_c \sum_{k'} h_{ck't}}. \quad (8)$$

For an ordered channel pair (c, j) in month t , content similarity is defined as

$$C_{cjt} = \sum_k \frac{p_{ckt} p_{jkt}}{q_{kt}} = \sum_k q_{kt} \left(\frac{p_{ckt}}{q_{kt}} \right) \left(\frac{p_{jkt}}{q_{kt}} \right). \quad (9)$$

The content measure is very similar to the audience-based definition: the content measure asks whether two channels overlap on content beyond what each feature’s overall popularity would predict. It equals one when the two channels overlap exactly as popularity predicts, and exceeds one when they share content that is distinctive to the two channels.

C.2 Data Collection

Audience similarity is built from the viewing records described in [Section 4](#) and requires no additional data. For each allocation event we build the audience similarity measure using the set of distinct logged-in users observed on each of the 100 channels over the 30 days preceding the event day.

Content is measured from a monthly scrape of the Twitch data aggregator SullyGnome covering September through December 2022.⁴ Each record gives, for one channel in one calendar month, the set of unique games streamed and the hours streamed per game. The sample covers 839 distinct games.

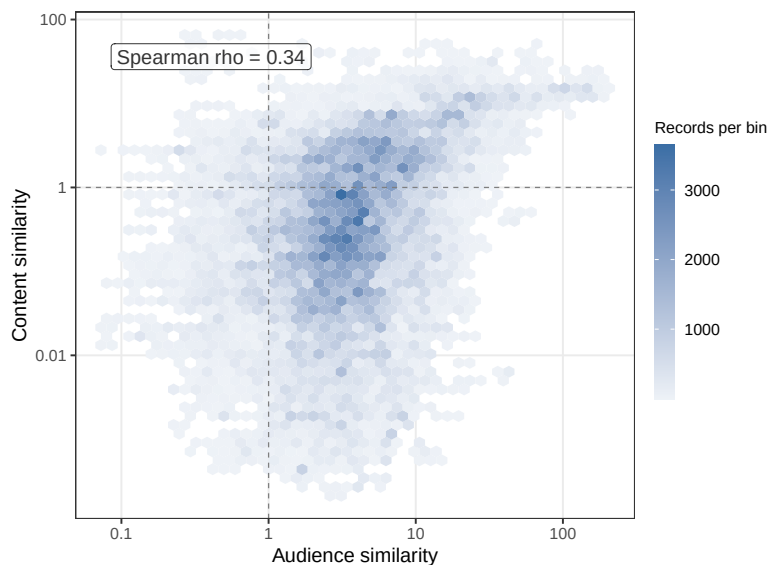
C.3 Comparing the Two Measures

Audience overlap is small in absolute terms but large relative to independence. The median pair of channels shares 0.9% of its combined audience, with a tenth percentile of 0.1% and

⁴<https://sullygnome.com/>

a ninetieth of 4.4%. The audience similarity (PMI) of the median channel pair is 2.6, and 87.1% of pairs exceed the independence threshold of one. The two measures agree moderately. [Figure C.1](#) plots them against one another over all cohorts. Their Spearman rank correlation is 0.342.

Figure C.1: Content Against Audience Similarity



Notes: Each observation is a cohort–other-channel pair of content and audience similarity. The dashed lines reference the independence value of one for each similarity measure. Shading indicates the number of observations per bin. $N = 621,802$.

D Robustness and Sensitivity

Our preferred results use double/debiased machine learning to estimate overlap-weighted average treatment effects for particular outcome definitions of watch, chat, subscribe, and platform behaviors. Because the identifying assumption of unconfoundedness cannot be tested directly, we assess the sensitivity of those estimates in four ways. We re-estimate the effects under alternative definitions of the outcome variables ([Online Appendix D.1](#)); we re-estimate them on alternative estimation samples, including a regression-discontinuity-style sample that exploits user-level channel join time relative to the timing of treatment ([Online Appendix D.2](#)); we re-estimate the effects with alternative linear and machine-learning models ([Online Appendix D.3](#)); and we quantify how strong an unobserved confounder would have to be to overturn each estimate ([Online Appendix D.4](#)).

D.1 Outcome Definitions

This subsection re-estimates the main effects under three alternative treatments of the outcome variables: the log and inverse hyperbolic sine transformations of the level outcomes, binary indicators for exceeding a sequence of watch and chat thresholds, and a narrower definition of what counts as a paid subscription.

D.1.1 Log and Inverse Hyperbolic Sine Transformations

Our main specification presents the watch and chat effects in levels (minutes and counts). A level effect can be misleading here for two reasons. First, consumption on the platform is heavily right-skewed, so a small number of heavy users can disproportionately influence the average level effects. Second, the platform may not equally value the same absolute increase for light and heavy users. For example, [Gentzkow et al. \(2024\)](#) show that the CPM (cost per thousand impressions) for heavy TV viewers is lower than that of lighter viewers, reflecting their reduced value to advertisers. We therefore report two transformations of

the level outcomes: the log, $\log(1 + Y)$, and the inverse hyperbolic sine (IHS), $\text{asinh}(Y) = \log(Y + \sqrt{Y^2 + 1})$, which behaves like the log for large values but is defined at zero.

Table D.1: Average Treatment Effects with Transformed Outcomes

Outcome	Channel	Months since treatment			
		M1	M2	M3	M4
Log watch	Trial	0.197 (0.004)	0.325 (0.008)	0.318 (0.009)	0.284 (0.008)
	Other	0.190 (0.008)	0.243 (0.009)	0.247 (0.009)	0.226 (0.009)
Log chat	Trial	0.365 (0.004)	0.145 (0.003)	0.128 (0.003)	0.115 (0.003)
	Other	0.102 (0.003)	0.102 (0.003)	0.096 (0.003)	0.086 (0.003)
Log sessions	Trial	0.166 (0.003)	0.164 (0.004)	0.156 (0.004)	0.126 (0.004)
	Other	0.113 (0.004)	0.132 (0.004)	0.131 (0.004)	0.115 (0.004)
IHS watch	Trial	0.200 (0.004)	0.352 (0.008)	0.347 (0.009)	0.312 (0.009)
	Other	0.207 (0.009)	0.264 (0.010)	0.269 (0.010)	0.248 (0.010)
IHS chat	Trial	0.436 (0.004)	0.174 (0.004)	0.155 (0.004)	0.138 (0.004)
	Other	0.122 (0.003)	0.122 (0.004)	0.115 (0.004)	0.103 (0.003)
IHS sessions	Trial	0.189 (0.003)	0.194 (0.004)	0.186 (0.005)	0.152 (0.004)
	Other	0.130 (0.005)	0.154 (0.005)	0.153 (0.005)	0.135 (0.005)

Notes: Overlap-weighted average treatment effects on $\log(1 + Y)$ (top panel) and $\text{asinh}(Y)$ (bottom panel). Cohort-clustered standard errors are in parentheses. All effects are statistically significant at $p < 0.001$.

In [Table D.1](#), we show the results. Relative to the levels specification in the main text, both the logged outcome and IHS transformation effects have the same sign and are more precisely estimated. The interpretation is also consistent with the level results: for example,

trial-channel watch time rises during the benefit period, and the increase is still present and statistically significant three months after the benefit period ends.

Still, these estimated effects should be interpreted with caution as they reflect a combination of extensive and intensive margin changes (Chen and Roth, 2024). We next estimate effects on the probability of exceeding different watch and chat thresholds.

D.1.2 Threshold Watch and Chat Outcomes

As an alternative to the level outcomes presented in the main text, here we present a sequence of binary outcomes that indicate whether a user’s watch time or chat count is at least a specified threshold.

On the trial channel during the benefit period, the greatest changes in watch time occur in the 4-32 hour range, corresponding to moderate levels of monthly viewership. The change at the 128-hour threshold is an order of magnitude smaller (0.003) than the peak (equivalent to greater than 4 hours of watch time per day). On other channels, the largest changes are at smaller thresholds, between 30 minutes and 4 hours during the benefit period. Beyond the 16 hour threshold the effects shrink steadily, from 0.017 at 32 hours to 0.004 at 128 hours, though they remain statistically distinguishable from zero. The effects are largest at low to moderate watch-time thresholds and smaller at the highest thresholds.

Table D.2: Watch Threshold Effects

Channel	Threshold: $\Pr(Watch \geq x)$								
	30 min	1 hr	2 hrs	4 hrs	8 hrs	16 hrs	32 hrs	64 hrs	128 hrs
Trial	0.011 (0.000)	0.019 (0.001)	0.030 (0.001)	0.042 (0.001)	0.053 (0.001)	0.052 (0.001)	0.039 (0.001)	0.017 (0.001)	0.003 (0.000)
Other	0.024 (0.001)	0.024 (0.001)	0.024 (0.001)	0.024 (0.001)	0.023 (0.001)	0.021 (0.001)	0.017 (0.001)	0.010 (0.001)	0.004 (0.001)

Notes: Overlap-weighted average treatment effects on the probability of meeting or exceeding the indicated watch-time threshold during the benefit month. Cohort-clustered standard errors are in parentheses.

Table D.3 shows similar treatment effects for chat threshold outcomes. Once again, the largest effects are for thresholds indicative of light to moderate levels of engagement. We also see that effects are driven by a mixture of intensive and extensive margin changes in behavior. The probability of sending at least one chat increased by 0.14 on the trial channel and 0.04 on other channels, showing an increase in the propensity to chat. The effects at higher chat thresholds are smaller but remain statistically significant.

Table D.3: Chat Threshold Effects

Channel	Threshold: $\Pr(\text{Chat} \geq x)$								
	1	2	4	8	16	32	64	128	256
Trial	0.136 (0.001)	0.112 (0.001)	0.093 (0.001)	0.076 (0.001)	0.059 (0.001)	0.043 (0.001)	0.027 (0.001)	0.016 (0.001)	0.007 (0.000)
Other	0.039 (0.001)	0.033 (0.001)	0.027 (0.001)	0.021 (0.001)	0.016 (0.001)	0.011 (0.001)	0.007 (0.000)	0.004 (0.000)	0.002 (0.000)

Notes: Overlap-weighted average treatment effects on the probability of sending at least the indicated number of chat messages during the benefit month. Cohort-clustered standard errors are in parentheses.

D.1.3 Paid and Prime Subscriptions

As a subsidiary of Amazon, Twitch offers special benefits to Amazon Prime customers. During the time of our data collection, users were able to subscribe for free to one channel of their choice each month by linking their Amazon Prime account to Twitch. In the average treatment effects presented in Section 6, we show the effect of temporary premium benefits on paid subscriptions. In Table D.4, we show the effect of temporary premium benefits on user propensity to subscribe using Prime benefits.

Similar to the negative short-run effect of premium benefits on paid subscriptions to the trial channel, treated users are also less likely to activate a Prime subscription on the trial channel. The Prime effect is small but negative in every month and decays towards zero, whereas the paid subscription effect is negative in month 2 before turning positive in months 3 and 4. In contrast, treated users show an approximately equal increase in propensity to

Table D.4: Paid and Prime Subscriptions

Outcome	Channel	Months Post			
		M1	M2	M3	M4
Paid	Trial	—	-0.007 (0.001)	0.016 (0.001)	0.089 (0.001)
	Other	0.049 (0.001)	0.058 (0.001)	0.040 (0.001)	0.034 (0.001)
Prime	Trial	-0.036 (0.000)	-0.020 (0.001)	-0.005 (0.001)	-0.005 (0.001)
	Other	0.047 (0.001)	0.041 (0.001)	0.035 (0.001)	0.032 (0.001)

Notes: Overlap-weighted average treatment effects on Paid and Prime subscription outcomes, by month since treatment. Paid effects reproduce [Table 2](#). Trial-channel Paid subscriptions in M1 are omitted because they include the allocated benefit. Cohort-clustered standard errors are in parentheses.

activate a Prime subscription on another channel as they do to purchase a subscription to another channel.

D.2 Estimation Sample

This subsection varies how the estimation sample is built. We first vary the watch-time filter used to remove bot-like activity. We then re-estimate the effects on an eligibility-discontinuity sample that leverages variation in join time at the boundary of the eligibility window.

D.2.1 Watch Time Filter

[Table D.5](#) shows the robustness of the main results to alternative filtering thresholds for removing users with extreme watch levels from the sample. Our main estimation sample uses a threshold of 21,600 minutes (equivalent to twelve hours per day) of watch time across all observed channels during the month prior to treatment. We consider two alternative thresholds at 14,400 minutes (eight hours per day on average) and 28,800 minutes (sixteen hours per day on average). Overall, the estimates are not sensitive to the threshold. Relative to the twelve-hour sample, the eight-hour sample raises the trial-channel watch effect from 212 to 215 minutes and the sixteen-hour sample lowers it to 202. The remaining effects are essentially unchanged.

D.2.2 Eligibility-Discontinuity Sample

We construct a regression-discontinuity-style sample that exploits user-level channel join time relative to treatment, using a narrow window around each premium benefit allocation. Only users present on the trial channel at the allocation time can be selected for the promotion, so a user who joins the channel shortly after the allocation is ineligible because of this timing rather than because of their preferences or engagement history. Join timing within a narrow window around the allocation is plausibly unrelated to potential outcomes, which yields a comparison group whose control status does not depend on how recipients are chosen among eligible users.

Table D.5: Average Treatment Effects for Alternative Watch Filters

		Maximum Average Daily Watch		
		8 hours	12 hours	16 hours
Watch	Trial	215.04 (7.63)	211.62 (7.64)	201.51 (7.70)
	Other	255.91 (32.95)	258.31 (33.29)	242.03 (33.85)
Chat	Trial	9.87 (0.37)	9.65 (0.37)	9.69 (0.37)
	Other	3.57 (0.38)	3.45 (0.38)	3.99 (0.40)
Subscribe	Other	0.048 (0.001)	0.049 (0.001)	0.050 (0.001)
Sessions	Trial	2.787 (0.073)	2.751 (0.073)	2.702 (0.072)
	Other	4.070 (0.320)	4.126 (0.325)	3.974 (0.331)
Variety	All	0.243 (0.011)	0.249 (0.011)	0.254 (0.011)

Notes: Overlap-weighted average treatment effects on M1 outcomes in separately constructed samples with the indicated cap on average daily pre-treatment watch time. The 12-hour column corresponds to the main estimation sample. Other eligibility rules are unchanged, and models are refit for each alternative sample. Cohort-clustered standard errors are in parentheses.

Within each cohort we define three groups of users based on the time they joined the trial channel relative to the allocation time. *Just-eligible users* are eligible users who joined the trial channel within a narrow window—15 minutes in our primary specification—prior to the allocation time. We split the just-eligible user pool into just-eligible treated users and just-eligible control users. *Just-ineligible users* are ineligible users who join the trial channel in a symmetric time window after the allocation time, and therefore could not have been selected for the promotion. Comparing the just-eligible treated users to the just-ineligible users isolates the effect of the promotion among users with nearly identical join

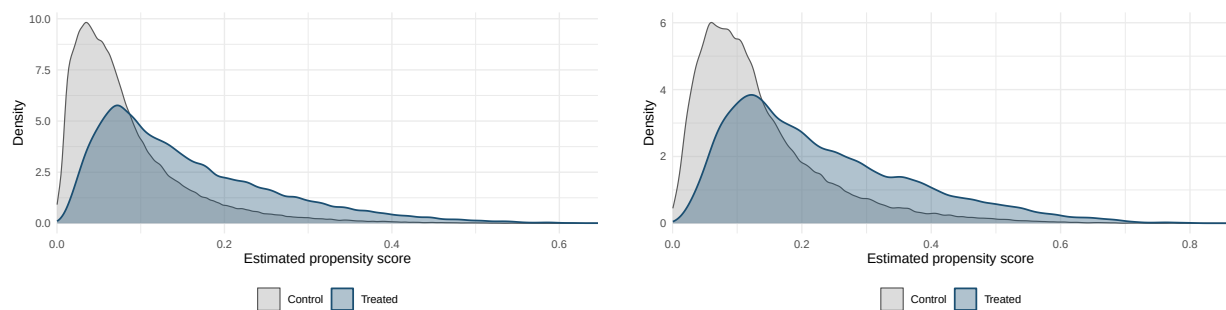
timing, while comparing the two control groups—just-eligible control users and just-ineligible users—provides a placebo test of the design.

Figure D.1 shows the resulting assignment-model propensity distributions: treated and comparison users overlap throughout, with no mass at degenerate propensities. The distributions are slightly more similar for the smaller bandwidth sample.

Figure D.1: Propensity Score Overlap, Eligibility-Discontinuity Sample

(a) 15-minute bandwidth

(b) 5-minute bandwidth



Notes: Propensity-score densities for just-eligible treated and just-ineligible users, using assignment models refit within the 15-minute and 5-minute bandwidth samples. Sample sizes (just-eligible treated/just-ineligible) are 26,652/269,139 and 11,534/64,854, respectively.

We estimate causal-forest overlap-weighted effects for these comparisons, retraining the nuisance models on each comparison sample. Table D.6 reports the two sets of estimates side by side. The left panel parallels the main results shown in Table 2, but compares just-eligible treated users to just-ineligible control users. The estimated effects are similar in sign and magnitude to our main results, and are directionally consistent. The right panel reports the placebo results. If there are no differences between just-eligible control users and just-ineligible control users, these columns should approach zero. We find a positive, but small, effect of being present for the allocation event. This increase in control user activity thus likely attenuates our main effects.

Table D.6: Eligibility-Discontinuity Sample: Causal and Placebo Contrasts

Behavior	Channel	Just-eligible treated vs. just-ineligible				Just-eligible untreated vs. just-ineligible (placebo)			
		M1	M2	M3	M4	M1	M2	M3	M4
Watch	Trial	158.82 (9.88)	93.86 (9.64)	75.58 (9.35)	52.96 (8.30)	16.56 (2.76)	11.77 (2.90)	9.10 (2.96)	9.36 (2.72)
	Other	141.12 (33.10)	123.36 (32.39)	120.01 (31.86)	87.22 (26.11)	4.11 (17.61)	-18.84 (18.83)	-40.50 (21.20)	-22.33 (18.15)
Chat	Trial	3.95 (0.60)	1.87 (0.71)	0.85 (0.67)	0.88 (0.57)	0.03 (0.17)	0.09 (0.21)	0.06 (0.20)	-0.38 (0.22)
	Other	0.41 (0.73)	0.33 (0.95)	0.62 (1.01)	0.21 (0.90)	0.36 (0.32)	0.51 (0.32)	0.36 (0.30)	0.00 (0.33)
Subscribe	Trial	—	-0.054 (0.002)	-0.006 (0.002)	0.060 (0.002)	—	0.001 (0.001)	0.002 (0.001)	0.002 (0.001)
	Other	0.047 (0.002)	0.050 (0.002)	0.033 (0.002)	0.031 (0.002)	0.000 (0.000)	-0.000 (0.001)	0.001 (0.001)	0.000 (0.001)
Sessions	Trial	—	1.050 (0.107)	0.790 (0.105)	0.612 (0.093)	—	0.101 (0.034)	0.093 (0.034)	0.050 (0.031)
	Other	1.855 (0.417)	1.464 (0.411)	1.103 (0.413)	0.951 (0.336)	-0.012 (0.178)	-0.066 (0.192)	-0.358 (0.229)	-0.325 (0.177)
Variety	All	—	0.057 (0.025)	0.067 (0.025)	0.075 (0.023)	—	-0.013 (0.006)	-0.003 (0.007)	-0.007 (0.007)
Engaged	All	—	0.007 (0.002)	0.010 (0.002)	0.013 (0.002)	—	-0.001 (0.001)	-0.002 (0.001)	-0.002 (0.001)

Notes: Causal-forest overlap-weighted average treatment effects for the indicated treated and placebo comparisons within 15 minutes of allocation. Columns show months since treatment. M1 Subscribe-Trial and Engaged are omitted because of allocated benefits and near-universal engagement, respectively. M1 Sessions-Trial and Variety are omitted as outcomes because they are near-sufficient predictors of eligibility in these narrow windows. Cohort-clustered standard errors are in parentheses.

The regression-discontinuity sample in [Table D.6](#) restricts each comparison to a 15-minute (three-pull) bandwidth on either side of the gift. As a robustness check, we re-estimate the same causal and placebo contrasts at a narrower 5-minute (one-pull; [Table D.7](#)) bandwidth. Tightening the window brings the compared groups closer in join timing at the cost of a smaller sample. The Other subscription effects remain close to the main estimates at the narrower bandwidth.

Table D.7: Regression-Discontinuity Sample at a 5-Minute (One-Pull) Bandwidth

Behavior	Channel	Just-eligible treated vs. just-ineligible				Just-eligible untreated vs. just-ineligible (placebo)			
		M1	M2	M3	M4	M1	M2	M3	M4
Watch	Trial	115.44 (13.96)	90.09 (14.08)	73.19 (13.70)	54.22 (11.85)	5.78 (4.93)	5.07 (4.98)	6.10 (5.10)	5.45 (4.78)
	Other	105.69 (49.38)	126.47 (47.28)	168.41 (45.23)	91.10 (35.08)	-6.53 (32.16)	-29.72 (38.33)	2.85 (35.47)	26.85 (39.36)
Chat	Trial	3.49 (0.93)	1.83 (0.80)	0.53 (0.85)	0.72 (0.73)	-0.08 (0.26)	-0.00 (0.27)	0.24 (0.37)	-0.24 (0.28)
	Other	-0.10 (1.07)	2.04 (1.22)	2.51 (1.33)	1.63 (1.21)	0.38 (0.41)	0.45 (0.50)	0.23 (0.55)	0.21 (0.53)
Subscribe	Trial	—	-0.050 (0.002)	-0.006 (0.002)	0.056 (0.003)	—	0.001 (0.001)	0.002 (0.001)	0.002 (0.001)
	Other	0.048 (0.003)	0.055 (0.003)	0.039 (0.003)	0.033 (0.003)	0.000 (0.001)	-0.001 (0.001)	0.002 (0.001)	0.002 (0.001)
Sessions	Trial	—	0.991 (0.164)	0.813 (0.157)	0.520 (0.135)	—	0.040 (0.059)	0.082 (0.060)	0.044 (0.055)
	Other	1.532 (0.554)	1.387 (0.530)	1.916 (0.568)	1.352 (0.439)	-0.070 (0.329)	-0.251 (0.325)	0.013 (0.330)	0.175 (0.292)
Variety	All	—	0.074 (0.036)	0.098 (0.036)	0.084 (0.034)	—	-0.004 (0.012)	0.008 (0.012)	0.001 (0.012)
Engaged	All	—	0.008 (0.002)	0.010 (0.003)	0.014 (0.004)	—	-0.002 (0.001)	-0.002 (0.001)	-0.003 (0.001)

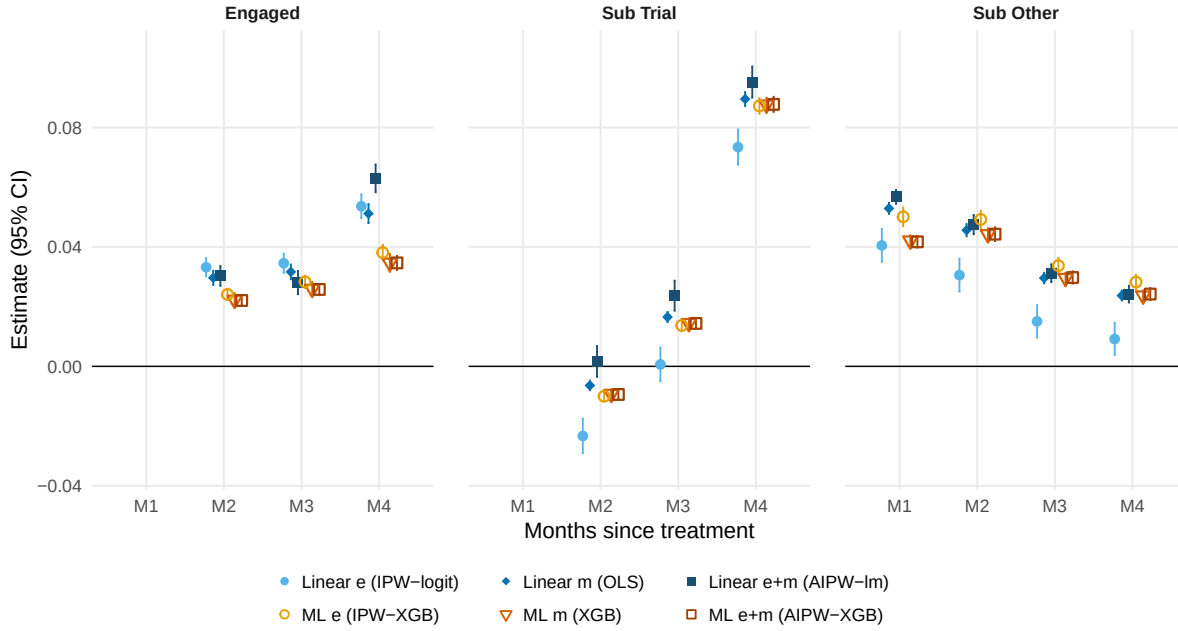
Notes: Causal-forest overlap-weighted average treatment effects for the indicated treated and placebo comparisons within 5 minutes of allocation. Columns show months since treatment. M1 Subscribe-Trial and Engaged are omitted because of allocated benefits and near-universal engagement, respectively. M1 Sessions-Trial and Variety are omitted as outcomes because they are near-sufficient predictors of eligibility in these narrow windows. Cohort-clustered standard errors are in parentheses.

D.3 Alternative Estimators

Next, we show how our estimates vary across six alternative specifications. Rather than re-estimating every outcome, we focus on the three binary outcomes used in the targeting analysis—trial-channel subscriptions, other-channel subscriptions, and Engaged—across the four months after treatment. All six specifications report the average effect on the treated, allowing outcome-only specifications to be compared with those that estimate a propensity model.

Specification 1 fits a linear model to control users' outcomes. Specification 2 uses inverse-probability weighting (IPW) with a logit assignment model. Specification 3 uses augmented inverse-probability weighting (AIPW), combining the outcome model of Specification 1 with the assignment model of Specification 2. Specification 4 estimates a gradient-boosted outcome model on control users, and Specification 5 uses IPW with a gradient-boosted assignment model. Specification 6 uses AIPW to combine the outcome model of Specification 4 with the assignment model of Specification 5. All models use the same pre-treatment covariates and cohort-level cross-fitting folds.

Figure D.2: Treatment Effect Estimates Across Models



Notes: Average treatment effects on treated users across six specifications. M1 Trial subscriptions and Engaged are omitted because they are mechanically affected by eligibility or allocation. Error bars are 95% confidence intervals from cohort bootstrap draws, holding fitted models fixed.

The estimates are shown in [Figure D.2](#). All six specifications yield estimates of the same sign and similar magnitude across all outcomes and months. The one exception is trial-channel subscriptions in Month 2, where effects are clustered around zero and one positive estimate is not statistically significant. The more flexible gradient-boosted specifications produce very similar estimates across all outcomes.

D.4 Sensitivity to Unobserved Confounding

Finally, we assess sensitivity to unobserved confounding using the omitted-variable-bias bounds for causal machine learning of [Chernozhukov et al. \(2026\)](#), which extend the framework of [Cinelli and Hazlett \(2020\)](#) to our DML estimator. We use observed covariate groups to benchmark how strongly an unobserved confounder would need to predict assignment and outcomes to reduce an estimated effect to zero.

We first measure the predictive strength of each covariate group in the assignment and outcome models. We then calculate how many times stronger the predictive strength of an unobserved confounder would need to be in both models to reduce the estimated effect to zero. This comparison allows the confounder to differ in its predictive strength across the two models, following patterns observed in the data.

We consider seven covariate groups: watch behavior, chat behavior, and subscription behavior on the trial and other channels separately, plus platform behaviors. When doing this analysis, we exclude groups that measure the same behavior as the outcome. For example, we exclude trial-channel watch behavior as a benchmark group for trial-channel watch outcomes. For Engaged, we exclude trial-channel watch and chat behavior.

[Table D.8](#) reports the smallest of these predictive strength multiples across the eligible benchmark groups, denoted k^* . For example, $k^* = 2$ means that a confounder with twice the selected group’s predictive strength in each model could reduce the estimate to zero. We report the smallest required multiple among the eligible benchmark groups. The group yielding this minimum, which we refer to as the binding group, can differ across outcomes and months. Note that, while these bounds describe the potential impact of unobserved confounding, they do not imply that such confounding is present.

Robustness to unobserved confounding differs across outcomes. The other-channel subscription effects are the most robust: reducing these estimates to zero would require a confounder with 8.2 to 12.0 times the predictive strength of the binding group, trial-channel chat behavior. Trial-channel subscription effects become more robust over time. The small

negative effect in Month 2 could be explained by a confounder with 0.41 times the strength of the binding group, whereas the Month 4 effect requires a multiple of 4.61. The multiples for Engaged range from 2.5 to 3.0.

The watch, chat, and session effects are more sensitive. For roughly half of the estimates, k^* is below one: a confounder weaker than the binding covariate group in both models could reduce the estimate to zero.

The three outcomes used in the targeting analysis in [Section 7](#) are comparatively robust. The Month 4 effects on trial-channel subscriptions, other-channel subscriptions, and Engaged require benchmark multiples of 4.61, 8.23, and 2.95, respectively, to be reduced to zero.

Table D.8: Sensitivity to Unobserved Confounding: Minimum Benchmark Multiple k^*

Outcome	Channel	Months since treatment			
		M1	M2	M3	M4
Watch	Trial	2.29 ^P	1.90 ^P	1.66 ^P	1.27 ^P
	Other	0.40 ^{WT}	0.45 ^{WT}	0.40 ^P	0.57 ^{WT}
Chat	Trial	0.75 ^{SO}	0.58 ^{SO}	0.25 ^{SO}	0.52 ^P
	Other	1.24 ^{CT}	0.54 ^{SO}	0.28 ^{SO}	0.27 ^{SO}
Subscribe	Trial	–	0.41 ^{SO}	1.08 ^{SO}	4.61 ^{SO}
	Other	11.98 ^{CT}	11.52 ^{CT}	9.82 ^{CT}	8.23 ^{CT}
Sessions	Trial	1.67 ^{WT}	1.91 ^{WT}	1.77 ^{WT}	1.60 ^{WT}
	Other	0.72 ^{WT}	0.99 ^{WT}	0.97 ^{WT}	0.94 ^{WT}
Variety	All	2.70 ^{WT}	3.42 ^{WT}	3.49 ^{WT}	4.36 ^{WT}
Engaged	All	–	2.63 ^P	2.52 ^P	2.95 ^P

Notes: Cells report the confounder strength needed to eliminate an effect, relative to an observed covariate group’s contributions to predicting assignment and the outcome. We report the smallest value across eligible groups. Superscripts identify that group: W = watch, C = chat, S = subscriptions; T/O = trial/other; P = sessions and variety. The outcome’s own behavior group is excluded; Engaged excludes trial watch and chat. – marks outcomes not estimated.

E Multi-Objective Targeting

The targeting analysis in [Section 7](#) uses causal forests to rank users and an IPW estimator to evaluate policy performance. [Online Appendix E.1](#) considers an alternative estimator for evaluating policies, holding the ranking model fixed, and [Online Appendix E.2](#) considers alternative models for ranking users, holding the evaluation method fixed.

E.1 Robustness of Policy Evaluation

As an alternative to the IPW estimator used to evaluate policy performance, in this section we evaluate policies using doubly robust (DR) scores ([Dudík et al., 2011](#); [Athey and Wager, 2021](#)). We fit separate gradient-boosted outcome models to treated and untreated users in the evaluation sample. Let $m_{jw}(X_i) = \mathbb{E}[Y_{ij} \mid W_i = w, X_i]$ denote the conditional expected outcome under treatment status $w \in \{0, 1\}$, and let $\hat{m}_{jw}(X_i)$ denote its estimate. As in the main text, we also fit a propensity model on the evaluation sample. Let $\hat{e}(X_i)$ denote the estimated propensity model. The DR score for user i and outcome j is

$$\hat{\psi}_{ij}^{\text{DR}} = \hat{m}_{j1}(X_i) - \hat{m}_{j0}(X_i) + \frac{W_i[Y_{ij} - \hat{m}_{j1}(X_i)]}{\hat{e}(X_i)} - \frac{(1 - W_i)[Y_{ij} - \hat{m}_{j0}(X_i)]}{1 - \hat{e}(X_i)}. \quad (10)$$

Each counterfactual policy’s performance is the average of these scores among the policy’s targeted users:

$$\hat{\rho}_j^{\text{DR}}(\pi) = \frac{1}{|T(\pi)|} \sum_{i \in T(\pi)} \hat{\psi}_{ij}^{\text{DR}}. \quad (11)$$

Here $|T(\pi)|$ is the number of targeted users. For observed allocation, we apply the normalized weights for the average treatment effect on treated users defined above to $Y_{ij} - \hat{m}_{j0}(X_i)$ rather than to Y_{ij} .

[Table E.1](#) compares IPW and DR policy performance estimates, holding the set of targeted users fixed across estimators. Each single-objective policy improves its focal outcome relative to random targeting under both the IPW and DR policy evaluators. Estimated mag-

Table E.1: Policy Values under Alternative Evaluation Estimators

Policy	Subscribe (Trial)		Subscribe (Other)		Engaged	
	IPW	DR	IPW	DR	IPW	DR
Max Subscribe (Trial)	11.87 (0.45)	11.98 (0.45)	3.80 (0.23)	3.51 (0.23)	4.75 (0.56)	3.82 (0.54)
Max Subscribe (Other)	8.31 (0.39)	7.55 (0.34)	10.90 (0.47)	11.01 (0.43)	3.17 (0.50)	3.21 (0.45)
Max Engaged	7.97 (0.50)	6.88 (0.35)	5.60 (0.47)	5.33 (0.32)	9.10 (0.90)	8.88 (0.63)
Equal Weights	10.74 (0.47)	9.79 (0.40)	9.70 (0.46)	9.35 (0.40)	7.96 (0.60)	5.70 (0.49)
Anchor 0.6 Subscribe (Trial)	11.50 (0.44)	11.32 (0.43)	7.22 (0.35)	7.31 (0.35)	3.36 (0.49)	2.95 (0.47)
Anchor 0.6 Subscribe (Other)	10.19 (0.43)	9.62 (0.41)	9.95 (0.43)	10.29 (0.42)	3.49 (0.46)	3.52 (0.44)
Anchor 0.6 Engaged	8.03 (0.45)	6.95 (0.33)	10.55 (0.56)	9.62 (0.41)	9.06 (0.74)	7.11 (0.52)
Observed	8.40 (0.23)	8.65 (0.22)	2.48 (0.23)	2.45 (0.19)	2.96 (0.23)	2.36 (0.20)
Random	8.28 (0.42)	7.64 (0.36)	5.10 (0.35)	4.94 (0.30)	6.35 (0.62)	5.25 (0.50)

Notes: M4 policy values in percentage points from inverse-propensity weighting (IPW) and doubly robust (DR) evaluation on the held-out policy sample. Cohort-clustered standard errors are in parentheses.

nitudes are largely consistent, although the estimated gains in Engaged relative to random targeting are smaller under DR evaluation for Equal Weights and the Engaged-anchored policy.

E.2 Alternative Ranking Models

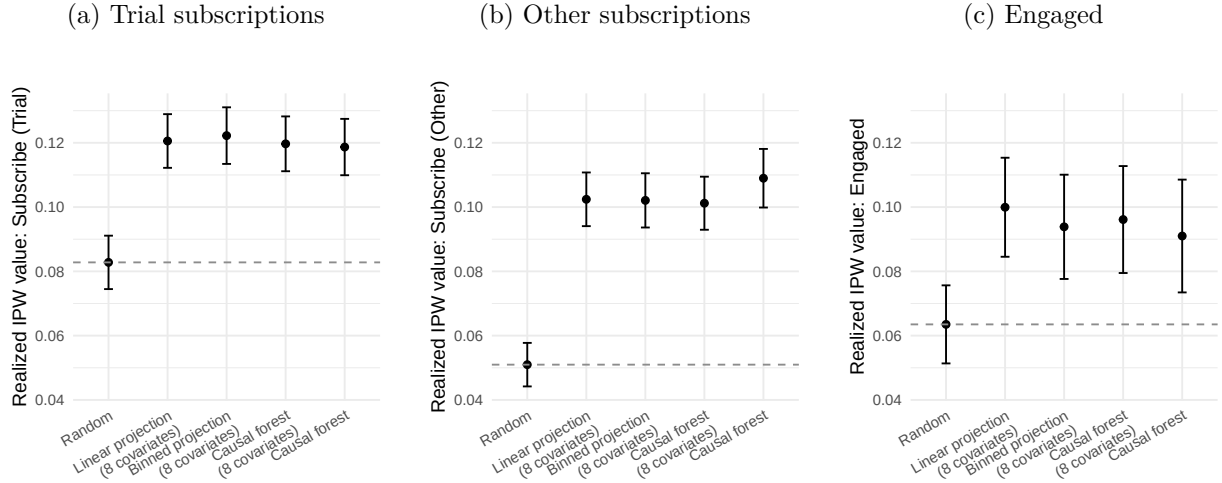
The policies in [Section 7](#) target the users with the largest weighted sums of causal forest estimates of conditional treatment effects. This subsection asks how much of the resulting targeting gain is retained by different conditional treatment effect estimators that summarize treatment effects with a smaller number of user characteristics, following the projection approach of [Semenova and Chernozhukov \(2021\)](#) and [Ellickson et al. \(2023\)](#).

For each outcome j , we first select the eight covariates with the highest variable importance in the estimation sample causal forest. We then compute, for each user in the estimation sample, the doubly robust score $\hat{\psi}_{ij}^{\text{DR}}$ analogous to [Equation 10](#), with nuisance models fitted on the estimation sample. Let $b_j(X_i)$ denote a vector of basis functions of the eight covariates selected for outcome j . We regress the DR scores on these basis functions by least squares to obtain

$$\hat{\gamma}_j = \arg \min_{\gamma} \sum_i \left(\hat{\psi}_{ij}^{\text{DR}} - b_j(X_i)' \gamma \right)^2, \quad (12)$$

where the sum runs over users in the estimation sample. The projected treatment effect is $\tilde{\tau}_j(X_i) = b_j(X_i)' \hat{\gamma}_j$. Because the doubly robust score evaluated at the true nuisance functions has conditional mean $\tau_j(X_i)$, $\tilde{\tau}_j$ estimates the best linear approximation to $\tau_j(X_i)$ in the span of the basis functions (Semenova and Chernozhukov, 2021). The projection estimates replace the forest estimates with $\tilde{\tau}_j$: each policy targets the users in each cohort with the largest weighted sums of projected effects. We consider two projections: a linear basis, in which continuous variables enter as $\log(1+x)$ and binary variables enter directly (nine parameters including the intercept), and a binned basis with indicators for up to eight quantile bins of each covariate.⁵ We compare the two projections with a causal forest whose splits are restricted to the same eight covariates and with the full forest.⁶

Figure E.1: Policy Performance by Ranking Model



Notes: Inverse-propensity estimates of each policy's effect on its focal M4 outcome. Intervals are 95% cohort-clustered confidence intervals.

⁵This results in 26 to 40 parameters, depending on the outcome. Since some covariates are binary (e.g., subscription activity), and others contain mass points at certain values (e.g., the mass point at zero chat activity), there are not eight distinct quantiles for every covariate.

⁶The restriction applies to the conditional effect model only; the assignment and outcome models used to construct the scores and to evaluate policies retain the full covariate set of the main text.

Figure E.1 reports the single-objective policy comparisons. The performance of all alternative conditional effect models is very similar from the full causal forest used in the main text.

Table E.2 reports the values of the seven policies of Section 7, with users ranked either by the forest estimates (Forest) or by the projected effects from the binned basis (Projection). The performance of the projection-based multi-objective policies is slightly worse than that of the forest-based multi-objective policies. Under projection-based policies, Equal Weights improves Trial and Other subscriptions relative to random targeting, by 2.7 and 3.1 percentage points, but the difference in Engaged relative to random targeting is statistically insignificant, whereas the forest-based policy improves all three outcomes.⁷ Low-dimensional projections therefore retain substantial single-objective targeting gains, but in our case multi-objective allocations are more sensitive to the policy model.

Table E.2: Policy Values under Alternative Ranking Models

Policy	Subscribe (Trial)		Subscribe (Other)		Engaged	
	Forest	Projection	Forest	Projection	Forest	Projection
Max Subscribe (Trial)	11.87 (0.45)	12.22 (0.45)	3.80 (0.23)	4.93 (0.26)	4.75 (0.56)	2.97 (0.49)
Max Subscribe (Other)	8.31 (0.39)	8.84 (0.40)	10.90 (0.47)	10.21 (0.43)	3.17 (0.50)	2.71 (0.44)
Max Engaged	7.97 (0.50)	8.61 (0.52)	5.60 (0.47)	3.41 (0.37)	9.10 (0.90)	9.39 (0.83)
Equal Weights	10.74 (0.47)	11.01 (0.46)	9.70 (0.46)	8.15 (0.40)	7.96 (0.60)	5.92 (0.56)
Anchor 0.6 Subscribe (Trial)	11.50 (0.44)	11.20 (0.44)	7.22 (0.35)	7.43 (0.33)	3.36 (0.49)	2.22 (0.43)
Anchor 0.6 Subscribe (Other)	10.19 (0.43)	10.38 (0.43)	9.95 (0.43)	9.09 (0.39)	3.49 (0.46)	2.23 (0.43)
Anchor 0.6 Engaged	8.03 (0.45)	8.78 (0.48)	10.55 (0.56)	7.14 (0.45)	9.06 (0.74)	8.04 (0.74)
Observed	8.40 (0.23)		2.48 (0.23)		2.96 (0.23)	
Random	8.28 (0.42)		5.10 (0.35)		6.35 (0.62)	

Notes: M4 policy values in percentage points from inverse-propensity weighting on the held-out policy sample, with users ranked by the forest estimates (Forest) or by the weighted sum of the three projected effects from the binned basis (Projection). Values are average effects among targeted users. Cohort-clustered standard errors are in parentheses.

⁷Ranking by the linear instead of binned projections leads to the same conclusion.